

- [Start](#)
- [Elementary Einstein](#)
 - [Special relativity](#)
 - [General relativity](#)
 - [Gravitational waves](#)
 - [Black Holes & Co.](#)
 - [Cosmology](#)
 - [Relativity and the Quantum](#)
- [Spotlights on relativity](#)
 - [Special relativity](#)
 - [General relativity](#)
 - [Gravitational waves](#)
 - [Black Holes & Co.](#)
 - [Cosmology](#)
 - [Relativity and the Quantum](#)
 - [Complete Spotlights](#)
- [About Einstein Online](#)
 - [The concept of EO](#)
 - [About Albert Einstein](#)
 - [About Us](#)
 - [Contributing Institutions](#)
 - [FAQ](#)
 - [Copyright policy](#)
 - [Contact](#)
- [Essentials](#)
 - [Links](#)
 - [Glossary](#)
 - [Further reading](#)
- 
- 



Glossary

More than 400 keywords from relativity and related topics, from “absolute zero” to “XMM Newton” – please use the menu on the left to choose a letter.

[A](#) | [B](#) | [C](#) | [D](#) | [E](#) | [F](#) | [G](#) | [H](#) | [I](#) | [J](#) | [K](#) | [L](#) | [M](#) | [N](#) | [O](#) | [P](#) | [Q](#) | [R](#) | [S](#) | [T](#) | [U](#) | [V](#) | [W](#) | [X](#) | [Y](#) | [Z](#)

absolute zero

The lowest possible temperature. It is the point where particles reach their minimum of vibrational motion. For instance, the temperature of a gas is proportional to the average kinetic energy of the moving molecules or atoms. In classical terms, absolute zero would be reached when the gas particles do not move anymore. According to quantum mechanics the particles still retain a minimum motion, induced by lowest possible energy state that a quantum mechanical system may have and which fluctuates according to the [Heisenberg uncertainty principle](#).

On the usual temperature scales, absolute zero is at -459.67 degrees Fahrenheit or -273.15 degrees [Celsius](#). By definition of the [Kelvin scale](#), absolute zero is at zero Kelvin.

abundances of elements

What portion of atomic matter in the universe is made up of hydrogen atoms? What fraction takes the form of helium, and how abundant are the other chemical [elements](#)? Questions like these are interesting in the context of relativity theory because the relativistic [big bang models](#) predict how many nuclei of light elements (mainly [deuterium](#), [helium](#), [lithium](#)) should have formed in the early universe during the phase known as [Big Bang Nucleosynthesis](#). A brief account of this phase can be found in the spotlight text [Big Bang Nucleosynthesis](#), while [Equilibrium and change](#) provides more information about the physical processes involved.

Measuring the abundances for these elements and subtracting the estimate for how many such nuclei formed inside stars (stellar nucleosynthesis) makes for an important test of this prediction and thus of the big bang models themselves. More information is provided by the spotlight text [Elements of the past](#).

acceleration

Every change of velocity with time is an acceleration.

This definition is slightly different from our everyday usage of the word. Ordinarily, we talk of an object accelerating when it becomes faster and faster. The physics definition covers two more situations. An object that decelerates, becomes slower, thus changes its velocity and, in the physics sense, undergoes a (negative) acceleration. Also, in physics, velocity is not the same as speed. A constant velocity implies not only constant speed, but also a constant direction of movement. Once the direction changes, so does the velocity – the change in velocity is associated with the change in the direction of movement. Thus, in the physics sense, even a car going around a curve of the road at constant speed undergoes acceleration.

accretion

When gas, dust or other kinds of matter fall towards a compact object (such as a [black hole](#) or a [neutron star](#)), a disk of infalling matter forms around the central object called the accretion disk.

The energy that matter gains in its fall is transformed into heat energy of the disk matter. Consequently, accretion disks are, as a rule, extremely hot. Their thermal

radiation they emit is an important tool for indirect observation of neutron stars and black hole.

Within the disk, matter spirals around and around, coming closer and closer to the central object until at last it falls onto its surface (or, in the case of a black hole, through its [event horizon](#)).

Synonyms: accretion disk

action-at-a-distance

Forces acting from one location to another without the need for any material connection, and without any delay – for instance, the Newtonian [gravitational force](#) with which even distant bodies in empty space can exert influence on each other.

active galactic nuclei (AGN)

The innermost regions of young galaxies can be very active and radiate considerable amounts of energy. Examples for such active galactic nuclei are radio galaxies and [quasars](#).

In current models, the energy source powering activities of such nuclei, is a [supermassive black hole](#) in the galactic centre.

age of the universe

Another word for [cosmic time](#), the time coordinate of the [big bang models](#): time as measured by clocks that are at rest relative to the expanding space, and that have been set to zero at the very beginning, the time of the hypothetical big bang singularity.

aggregate state

See under [states of matter](#).

Albert Einstein Institute

One of the research institutes of the [Max Planck Society](#); an international centre for research on Einstein's theory of gravity – from the mathematical fundamental, astrophysics and gravitational waves to quantum gravity. Founded in 1995, the institute is situated in Golm near Potsdam in Germany. In 2002, the experimental branch of the institute was opened in Hannover. It is dedicated to research with the gravitational wave detector [GEO 600](#) .

[AEI webpages](#)

[Website of AEI-Hannover](#)

Synonyms: Max Planck Institute for gravitational physics AEI

alpha particle

Alternative expression for the naked (i.e. stripped of electrons) atomic nucleus of the element helium consisting of two protons and two neutrons.

amplitude

For a physical quantity that changes periodically, the amplitude is a measure of how much the quantity changes from maximum to minimum. The simplest example is a sine oscillation. Over time, the sine curve oscillates between its minimum and its maximum values and the amplitude measures how big this oscillation is. There are different ways of defining amplitude. Some definitions use the Peak-to-peak difference for the amplitude (Maximum of the signal minus Minimum of the signal). Other definitions for signals with values centered symmetrically around zero specify the amplitude as the maximum value of the signal (Half of the Peak-to-peak amplitude).

Depending on the nature of the oscillation or wave, the amplitude will have different meanings. For a pendulum swinging back and forth, the amplitude is the maximum angle between the vertical direction and the pendulum string. For an electromagnetic wave, the amplitude is the maximal value of the electric field or equivalently (since the two maxima are related) the maximum of the magnetic field. For a (weak) gravitational wave, the amplitude is a direct measure of the changes in distance caused by the wave – as a simple gravitational wave of amplitude A passes, there are two directions in which distances are alternately stretched by up to a factor $(1+A/2)$ and compressed by a factor $(1-A/2)$.

The amplitude can change over time. For instance, for an ordinary pendulum, air friction will slow the pendulum bob down, and for each period – for each time the pendulum bob travels back and forth – the amplitude will be less than for the previous period. For a wave, the amplitude will also in general vary with location. Typically, the amplitude of a wave will decrease with the distance from the wave's source.

angular momentum

A [conserved](#) physical quantity associated with the rotation of an object.

In [classical](#) physics, the contribution of each part of a body to the body's total angular momentum is the part's mass times its distance from the axis of rotation, times the part's speed component which points in the direction of rotation and is perpendicular to the axis of rotation.

More information can be found in the spotlight text [What figure-skaters, planets, and neutron stars have in common](#).

In the context of [general relativity](#), angular momentum is an interesting quantity in the physics of [black holes](#). More information about this can be found in the spotlight text [How many kinds of black holes are there?](#)

anti-particle

As a general rule, theories uniting [special relativity](#) and [quantum theory](#) predict the existence of a species of anti-particle for every species of particle. For instance, if such a theory contains [electrons](#), then it also contains their anti-particles, called [positrons](#), for [protons](#), there are anti-protons, and so on.

It is a universal feature of anti-particles that they have the same [mass](#) as corresponding particles, and equal, but opposite [charges](#); for examples, electrons and positrons have the same mass, but the electrons carry negative electric charge,

whereas positrons carry the exact same amount of positive electric charge. For particles that carry no charges of any kind, particles and anti-particles are identical.

Synonyms: anti-matter, anti-particles

aphelion

For a [planet](#) or other heavenly body orbiting the [sun](#) on an [elliptic orbit](#), that point of the orbit farthest from the sun. The point closest to the sun is the [perihelion](#). In the context of [general relativity](#), aphelion and perihelion are of great interest as that theory predicts a slight motion of these points around the sun, cf. [\(relativistic\) perihelion shift](#).

arcminute

Synonyms: minute of arc, second of arc. Subdivisions of an angle, analogous to subdivisions of time: Sixty arcseconds correspond to one arcminute; sixty arcminutes (or 3600 arcseconds) correspond to one degree. A right angle has 90 degrees, or 5400 arcminutes, or 324000 arcseconds.

To denote fractions of these units, a prefix is added in the usual way – for instance, one thousandth of a second of arc is a [milli](#)second of arc.

Synonyms: arcsecond

astronomical unit

Unit of length used by astronomers for distances in and around the [solar system](#); the average distance from the earth to the sun. Abbreviation: au.

Concrete value:

1 au = 149.597870700 million kilometres
= 92.955807 miles
= 8.3 [light minutes](#).

atom

All matter we encounter in everyday life consists of smallest units called atoms – the air we breath consists of a wildly careening crowd of little groups of atoms, my computer's keyboard of a tangle of atom chains, the metal surface it rests on is a crystal lattice of atoms. All the variety of matter consists of less than hundred species of atoms (in other words: less than a hundred different [chemical elements](#)).

Every atom consists of an [nucleus](#) surrounded by a cloud of [electrons](#). Nearly all of the atom's [mass](#) is concentrated in its nucleus, while the structure of the electron cloud determines how the atom can bind to other atoms (in other words: its chemical properties). Every chemical element can be defined via a characteristic number of [protons](#) in its nucleus. Atoms that have lost some of their usual number of electrons are called [ions](#). Atoms are extremely small (typical diametres are in the region of tenths of a billionth of a metre = 10^{-10} metres), and to describe their properties and behaviour, one has to resort to [quantum theory](#).

atomic nucleus

See [nucleus](#).

Auger observatory

See [Pierre Auger Observatory](#).

baryon

In [particle physics](#): collective expression for particles that fundamentally consist of three [quarks](#). The most important examples are [protons](#) and [neutrons](#).

In [cosmology](#), the word denotes ordinary matter in contrast with more exotic forms of matter that the greatest part of [dark matter](#) is thought to consist of. This usage comes about because cosmologists are mainly interested in what percentage of mass in the universe is represented by ordinary matter. The mass of ordinary matter is mostly contained in atomic [nuclei](#), and these nuclei are built of baryons ([protons](#) and [neutrons](#)), the total baryonic mass is, to good approximation, the same as the total mass of ordinary matter in any region of our universe.

big bang

The big bang models are the foundation of modern [cosmology](#). Firmly grounded in Einstein's theory of [general relativity](#), they describe a universe that began in a very hot initial state and has expanded (and cooled down) ever since. They make precise predictions about [nucleosynthesis](#) in the early universe, the existence and properties of the [cosmic background radiation](#), and the distribution of distant [galaxies](#) in the cosmos, which have been confirmed by astronomical observation.

The word “big bang” has two different meanings. In a strict sense, the big bang is a [spacetime singularity](#), a state of infinite [density](#) – the initial state the big bang models predict for our universe. In a more general sense, the term is applied to the earliest cosmic eras, in which the universe was exceedingly hot and dense. Further information about these two meanings and why it is important to distinguish between them can be found in the spotlight text [A tale of two big bangs](#).

The basic features of the big bang models are reviewed in the chapter [Cosmology of Elementary Einstein](#).

Selected aspects of cosmology are described in the category [Cosmology](#) of our [Spotlights on relativity](#).

Big Bang Nucleosynthesis

Synonym: primordial [nucleosynthesis](#). The formation of complicated nuclei from constituents such as protons and neutrons in the early universe. According to the [big bang models](#), the early universe was filled with a particle soup of [protons](#) and [neutrons](#). At [cosmic times](#) between a few seconds and a few minutes, nuclear reactions produced the first light elements, mainly nuclei of [deuterium](#), different varieties of [helium](#) and [lithium](#). Heavier nuclei up to those of iron formed and continue to form in the course of [fusion](#) processes inside [stars](#); nuclei that are even more massive form in the course of [supernova explosions](#). These explosions also serve to disseminate the complex nuclei formed inside stars (stellar nucleosynthesis) in space.

A brief account of Big Bang Nucleosynthesis can be found in the spotlight text [Big Bang Nucleosynthesis](#), while [Equilibrium and change](#) provides more information about the physical processes involved and [Elements of the past](#) describes how the predictions of Big Bang Nucleosynthesis can be tested against astronomical observation.

binary star

A system consisting of two stars in orbit around each other. From a relativistic point of view, there are binaries that are of special interest, namely those in which at least one of the partners is a [neutron star](#) or a [black hole](#). Potentially, such systems are effective sources of [gravitational waves](#).

Synonyms: binary

binding energy

The [energy](#) needed to break up a composite object into its component parts.

More about binding energy, the [mass defect](#) it is responsible for and its role in [nuclear fission](#) and [nuclear fusion](#) can be found in our spotlight topic [Is the whole the sum of its parts?](#)

Birkhoff's theorem

A theorem of [general relativity](#), discovered by J. T. Jebsen (1921) and independently discovered, and named after, George D. Birkhoff (1923): Any spherically symmetric spacetime has the same properties as some region from one of a simple family of spacetimes found by Karl Schwarzschild in 1916. More concretely: The spherically symmetric spacetime around any spherically symmetric matter configuration (approximately, in good approximation, the earth) has the same properties as spacetime around a [Schwarzschild black hole](#) of the appropriate mass.

BKL conjecture

Conjecture by the Soviet physicists Vladimir Belinskii, Isaak Khalatnikov and Evgeny Lifshitz that, near a [singularity](#), the contribution of [matter](#) to [gravity](#) becomes negligible compared with the effects of gravity as a source of further gravity (compare the spotlight text [The gravity of gravity](#)), and that near a singularity, the variation of the gravitational field from one location to the next can be neglected – what is much more important is the way gravity changes over time. Further information about this can be found in the spotlight text [Of singularities and breadmaking](#).

black hole

Region in space where a sufficient amount of mass is concentrated so that it forms a gravitational prison – a region into which matter or light can enter from the outside, but from which nothing that has once entered can ever leave.

Basic information on this key phenomenon of Einstein's [general theory of relativity](#) can be found in the chapter [Black holes & Co.](#) of [Elementary Einstein](#).

Selected aspects of the physics of black holes and neutron stars are described in the category [Black holes & Co.](#) of our [Spotlights on relativity](#).

In Einstein's theory, black holes are truly black, due to the fact that no radiation or light can ever escape them. Once [quantum theory](#) is taken into account, this assumption no longer holds – on the contrary, it seems as if black holes should emit so-called [Hawking radiation](#). However, for astrophysical black holes (that typically have more or even much more than one [solar mass](#)), that radiation would be undetectable, even if we could transport today's most sensitive sensors into the immediate vicinity of the black hole.

Synonyms: black holes

black hole perturbation theory

Black hole perturbation theory is an approach to calculating theoretical gravitational waveforms, so-called [templates](#). It assumes a binary system consisting of a [black hole](#) and a compact small object which is treated as point particle perturbing [spacetime](#). The system thus qualifies as an [extreme mass-ratio inspiral \(EMRI\)](#).

black hole uniqueness theorems

Theorems proved in the context of [general relativity](#) that answer the question: How many different kinds of [black holes](#) are there? If that question is restricted to [stationary](#) black holes (namely black holes that have settled down and do not change over time), then the answer is: Surprisingly few. Once you know a stationary black hole's [mass](#), [angular momentum](#) (roughly speaking, how fast it rotates) and [electric charge](#), its properties are determined completely.

More information can be found in the spotlight text [How many different kinds of black hole are there?](#)

blackbody

Idealized body that is capable of absorbing and emitting all forms of [electromagnetic radiation](#), regardless of their [wavelength](#). The [thermal radiation](#) emitted by such a body is governed by a set of especially simple laws, like [Planck's radiation law](#), the [Stefan-Boltzmann law](#) and [Wien's law](#).

blueshift

The frequency of a simple [light wave](#) is directly related to its colour (cf. [spectrum](#)). For the highest possible frequencies of visible light, the colour is blue-violet. If the frequency of a light wave is shifted towards higher frequencies (for instance by the [doppler shift](#)), that corresponds to a colour shift towards the blue-violet end of the spectrum, and is hence called a blueshift.

In the light of that, the term 'blueshift' has acquired a more general meaning. It is used to refer to any shift towards higher frequencies, including types of electromagnetic radiation where the frequencies do not correspond to any visible colour, and more generally to other types of waves (for example, [gravitational waves](#)).

See also [redshift](#).

BNL

See [Brookhaven National Laboratory \(BNL\)](#).

boson

Collective expression for [quantum particles](#) with an integer [spin](#), mainly spin 0, spin 1, spin 2.

Among the [elementary particles](#), bosons are [carrier particles](#) in charge of transmitting the influences of forces. [Photons](#), for instance, the carrier particles of the electromagnetic force, are bosons. In contrast, the elementary particles that make up matter, such as [electrons](#) or [quarks](#), are so-called [fermions](#).

brane

In [string theory](#): An object that is the analogue of a two-dimensional membrane embedded in three-dimensional space – an entity with a certain number of [dimensions](#) (one-brane, two-brane, three-brane...) embedded in the higher-dimensional space of string theory. A one-brane or 1-brane has one spatial dimension, a two-brane has two, and so on.

brane world

The notion that our world with its three [dimensions](#) of [space](#) is a three-brane embedded in a higher-dimensional space, akin to a two-dimensional surface embedded in ordinary three-dimensional space.

More information about this can be found in the spotlight text [The embedded universe](#).

Brookhaven National Laboratory (BNL)

National laboratory in the United States, located on Long Island, New York. The BNL operates the [Relativistic Heavy Ion Collider \(RHIC\)](#), a [particle accelerator](#) that enables researchers to recreate the state of matter fractions of a second after the [big bang](#). Also, BNL operates accelerators used to produce [synchrotron radiation](#).

> [Brookhaven National Laboratory Website](#)

> [National Synchrotron Light Source II at the Brookhaven National Laboratory](#)

brown dwarf

A failed [star](#): A gas ball in space that has between one and ten percent [solar mass](#) – not enough for the temperature and pressure in its core to reach the values required for the [nuclear fusion](#) to start that would transform the gas ball into a shining star.

California Institute of Technology (Caltech)

Major research university located in Pasadena, California. Areas of research include [general relativity](#), [particle physics](#), [quantum gravity](#) and [cosmology](#); in addition, Caltech is one of the main sites for the researchers (although not the detectors!) of the [LIGO](#) project, which operates the most sensitive [gravitational wave detectors](#) to date. Also, the [Einstein Papers Project](#), which is working on an edition of Einstein's collected papers, is based at Caltech.

- > [Caltech webpages](#)
- > [Webpages of the Theoretical Astrophysics and Relativity group at Caltech](#)

carrier particle

In the framework of [relativistic quantum field theories](#) (which form the theoretical basis of the physics of [elementary particles](#), the forces by which matter particles interact are transmitted by so-called carrier particles travelling back and forth between them. For instance, the [electric force](#) between two [electrons](#) would come about through the exchange of [photons](#), the carrier particles of the [electromagnetic interaction](#). Carrier particles always have integer [spin](#), such as spin 1 or 2 (which means they belong to the class of particles called [bosons](#)). Synonym: force particles.

causal

In the context of relativity: causality concerns the question of which events cause which other events (Latin *causa*, the reason, the cause) or, more generally, can influence them. In [special relativity](#), nothing, no moving object, no information, no influence can move faster than light. Thus, in principle, an event can only influence another event if the hypothetical influence (such as a signal or a force) would not have to be transmitted faster than light. In other words, light propagation determines the causal structure of spacetime (cf. [light-cone](#)). Models and theories that take this structure into account are called causal – for example the [relativistic quantum field theories](#).

In [general relativity](#), the cosmic speed limit, the speed of light is only defined locally: In a side-by-side race, no object, no influence can overtake a light signal. Also from this a causal structure can be derived and determines which events can influence which other events. As [gravity deflects](#) and [delays](#) light signals, the matter is more complicated in general relativity than in special relativity. Although this makes the analysis somewhat more complicated, the following still applies: the propagation of light determines the causal structure.

Synonyms: causality causal structure

causal sets

Approach to the problem of finding a theory of [quantum gravity](#): In causet models, spacetime is composed of elementary building blocks that represent elementary [events](#).

More information about causal sets can be found in the spotlight topic [From order to Geometry: causal sets](#).

Synonyms: causets

celestial mechanics

That discipline in physics (and astronomy) dealing with the laws that govern the motions of heavenly bodies. Originally seen as distinct from the motions of bodies on earth (see [Kepler's laws of motion](#)), it has been a sub-discipline of [mechanics](#) ever since Newton derived the cosmic laws of motions from more general mechanical laws.

For most astronomical applications, Newtonian, [classical](#) mechanics works perfectly well, however, as soon as high-precision measurements or strong gravitational fields come into play, celestial mechanics is governed by laws of [relativistic mechanics](#) derived from Einstein's theory of [general relativity](#).

Celsius

Most European countries use the Celsius temperature scale in everyday life. Temperatures are given in “degrees Celsius” (abbreviated as °C). By definition, the zero point of this scale (0°C) is the melting point of water, while the temperature 100°C corresponds to its boiling point (both parts of the definition assume the same standard value for air pressure).

Relation to the [Fahrenheit scale](#): X degrees Celsius correspond to $(9/5 \text{ times } X) + 32$ Fahrenheit, Y Fahrenheit are $(Y - 32) * 5/9$ degrees Celsius.

Relation to the [Kelvin temperature scale used in physics](#): X degree Celsius are X plus 273.15 Kelvin, Y Kelvin are Y minus 273.15 degrees Celsius. In particular, differences in temperature are the same in Kelvin and in degrees Celsius; the only difference between the two scales is their choice of zero point.

censorship

See [cosmic censorship](#).

centrifugal force

An [inertial force](#) which an [observer](#) in a rotating [reference frame](#) needs to introduce in order to explain why nearly all objects in the vicinity appear to undergo [acceleration](#) away from the axis of rotation.

CERN

European research centre for nuclear and particle physics (Centre Européen pour la Recherche Nucleaire – pardon my French), located near Geneva on both sides of the franco-swiss border, founded 1954.

CERN isn't famous just because of particle accelerators like its proton synchrotron, the Large Electron Positron Collider (LEP) and the [Large Hadron Collider \(LHC\)](#), but also as the birthplace of the World Wide Web.

> [CERN website](#)

Chandrasekhar mass

Upper bound for the masses of [white dwarfs](#), in other words: for what low-mass [stars](#) become when they have used up their nuclear fuel. The first to calculate this upper bound was the Indian-American astrophysicist Subramanian Chandrasekhar.

The Chandrasekhar mass is 1.4 times as large as the [solar mass](#). The reason that no white dwarf can have more mass follows from its need to maintain equilibrium between the gravitational force working towards further collapse and the interior pressure of the star acting to prevent collapse. For larger masses, the [degeneracy pressure](#) on which a white dwarf's stability depends is overcome by the gravitational force, and further collapse ensues.

Synonyms: Chandrasekhar limit

charge

A measure of the strength of a force ([action-at-a-distance](#)) originating from a body, and of how susceptible it is to being influenced by other bodies via the same force. The most famous example is electric charge: Electrically charged bodies act on other electrically charged bodies via an [electric force](#) whose strength is proportional to the electric charges of the bodies involved.

It is a characteristic property of charges that they are conserved; they can neither be created from nothing nor simply disappear. For instance, when a [positron](#) with electric charge +1 (in suitable units) and an [electron](#) with electric charge -1 annihilate to give electromagnetic radiation, overall charge conservation is satisfied: Before the annihilation, the sum of the electric charges was $1+(-1)=0$, and afterwards, when there is only uncharged electromagnetic radiation left, it is also zero.

In the context of [particle physics](#), there are more abstract charges not directly connected with forces and interaction, but subject to similar conservation laws.

chemical element

See [element](#)

chemical evolution

The changes in the abundances of the different chemical [elements](#) that have taken place throughout the history of the universe, mainly in the very early universe during the phase called [Big Bang Nucleosynthesis](#) and, from a couple of hundred million years later until today, in the interior of [stars](#) (stellar [nucleosynthesis](#)).

classical

In physics, the word is used with two meanings. First of all, it denotes physical models or theories that take into account neither the effects of Einstein's [theories of relativity](#) nor those of [quantum physics](#), for example classical [mechanics](#). However, it is also used to denote models or theories that are not formulated in the framework of [quantum physics](#); in that sense, [general relativity](#) is an example for a classical theory.

classical mechanics

See [mechanics, classical](#)

classical tests of general relativity

See [tests of general relativity, classical](#)

cluster of galaxies

See [galaxy cluster](#)

Synonyms: clusters of galaxies

Compton Gamma Ray Observatory

A satellite observatory for astronomical observations of [gamma rays](#) operated by [NASA](#) from 1991 to 2000. Scientific aims included the study of [gamma ray bursts](#), [pulsars](#), [supernovae](#), and [accretion](#) processes around [black holes](#).

[CGRO homepage \(NASA\)](#).

conic singularity

A particular type of [spacetime singularity](#) (i.e. a boundary where [spacetime](#) ends) that is not associated with [curvature](#), but instead is the analogue of the pointed tip of a cone.

More information about the different types of singularity can be found in the spotlight text [Spacetime singularities](#).

conservation laws

Some of the most important quantities in physics are *conserved* : What they measure can neither be created nor destroyed, and their total sum is constant over time. Such statements of constancy over time are called *conservation laws*.

The most important conserved quantity is [energy](#). Energy can neither be created from nothing nor simply vanish. If the energy contained in a system increases, it must be because energy has been transported into the system (and there is now less energy outside the system); if the energy decreases, it must be because energy has been transferred from the system (and there is now more energy outside).

Another important class of conserved quantities are [charges](#), for instance [electric charge](#). A further conserved quantity in [mechanics](#) is [angular momentum](#); a quantity associated with a body's rotation.

Synonyms: conserved quantities

constancy of the speed of light

One of the basic postulates of [special relativity](#): The [speed of light](#) in a vacuum is the same for all observers drifting through gravity-free space (more precisely: for all [inertial observers](#)). In particular, its value is independent of an observer's motion relative to the source of the light.

continuous

Space as we are used to thinking about it is a continuum or, equivalently, continuous space: Between every two points, there always exists an infinity of other points, and every volume can be divided into smaller and smaller parts without ever reaching a limit.

Synonyms: continuum

coordinates

A rule for assigning to each point of a general [space](#) or [spacetime](#) a set of numbers for purposes of identification.

Many readers will know two examples from school: In the case of the line of real numbers, every point on the line corresponds to a real number which can be seen as its coordinate. What's important is that these coordinates reflect neighbourly relations: The number 1 lies between the number 0 and the number 2, and so does the point corresponding to it lie between the two points corresponding to 0 and 2. The second example is the usual X-Y-coordinate system (sometimes called Cartesian coordinates), by which every point in a plane can be characterized by two numbers: the first its X coordinate value, the second its Y coordinate value.

The examples reflect an important property of coordinates: To uniquely identify a point in space, one needs as many coordinate values as the space has [dimensions](#).

Of the four coordinates defining an event in [spacetime](#), three serve to fix its location in three-dimensional space, while the fourth gives the point in time for the event.

Synonyms: coordinate system

Coriolis force

A [fictitious](#) or [inertial force](#) which an [observer](#) in a rotating [reference frame](#) needs to introduce in order to explain why certain moving objects appear to undergo [acceleration](#) at right angles to their direction of motion.

The Coriolis force plays an important role in meteorology – from the point of view of an observer at rest on the surface of the earth, it explains the deflection of certain wind flows.

cosmic background radiation

“Electromagnetic echo” of the early universe; first predicted by the [big bang](#) models in the context of [general relativity](#); later, from the 1960s on, observed with radio telescopes.

The cosmic microwave background contains important information about the properties and the earliest history of the universe. For instance, it can be used to deduce whether space is [curved](#) or [Euclidean](#); more information about this can be found in the spotlight text [Cosmic sound](#).

cosmic censorship

Possibly the most disturbing feature of Einstein's [general theory of relativity](#) is the existence of [singularities](#) – most commonly, regions of [spacetime](#) in which [density](#) and [curvature](#) go to infinity.

It is quite likely that singularities are artefacts resulting from the fact that Einstein's theory does not take [quantum effects](#) into account, and that they will be absent in a more complete theory of [quantum gravity](#). Yet even if you leave aside quantum theory, and stay strictly within the framework of Einstein's theory, it is likely that most singularities are, if not absent, then at least well-concealed:

The hypothesis of cosmic censorship states that, whenever a body collapses so completely as to result in the formation of a singularity, a [black hole](#) will be formed so that the singularity will be hidden behind the [horizon](#), and thus completely unobservable for anyone outside the black hole.

At the present time, this hypothesis is unproven. Indeed, there are some counterexamples, but they describe idealized situations which are not likely to tell us anything about the real world. Finding a proof that, for all realistic collapse situations, there is indeed cosmic censorship, is one of the great open problems of general relativity research.

Cosmic Explorer

Cosmic Explorer is a proposed [third generation gravitational wave detector](#), which, together with other third generation detectors like the [Einstein telescope](#), is supposed to allow for [gravitational wave](#) detection with greater sensitivity than ever before. The detector will be built in the United States and is designed with an L-shaped geometry, each arm being 40 kilometers in length.

[Website of the Cosmic Explorer](#)

Cosmic microwave background radiation

See [cosmic background radiation](#), above.

cosmic rays

Highly energetic particles reaching the earth from the depths of space, mainly consisting of [protons](#) and light atomic [nuclei](#).

cosmic time

Measure for the progress of the evolution of an expanding universe such as that of the [big bang models](#). It corresponds to time as measured by clocks that are at rest relative to the expanding space, and that have been set to zero at the very beginning, the time of the hypothetical big bang singularity. Synonym: Age of the universe.

The basic features of the cosmological models that use cosmic time are reviewed in the chapter [Cosmology](#) of [Elementary Einstein](#).

cosmological constant

In the [big bang models](#), an inherent tendency of space to accelerate or decelerate its expansion. From observations, it seems that our own cosmos has a cosmological constant that leads to a slight acceleration of its expansion.

cosmological redshift

Consequence of cosmic expansion in the [big bang models](#): the farther away a galaxy, on average the more strongly shifted towards lower frequencies is the light we receive from it .

cosmology

That branch of physics and astronomy dealing with the structure and development of the universe as a whole. At the core of modern cosmology are the [big bang](#)

[models](#) based on Einstein's [general theory of relativity](#). Their basic features are reviewed in the chapter [Cosmology](#) of [Elementary Einstein](#). In order to describe the very early universe, it will be necessary to take the effects of [quantum gravity](#) into account – this gives rise to models of [quantum cosmology](#).

current

Matter in coordinated, flowing motion – think of water flowing in a pipe. An important example is the electric current associated with moving [electric charges](#). Electric currents are the sources of [magnetic fields](#).

curvature

For a two-dimensional [surface](#): criterion that allows us to decide whether that surface is a [plane](#), or part of a plane (i.e. a surface on which the usual rules of high school geometry apply), or not. Two possibilities to define the curvature of a plane are the following:

Sum of the angles of a triangle. In a plane, the sum of the three angles in a [triangle](#) formed by three straight lines is always 180 degrees. In a more general surface, the sum of the angles of a more general triangle formed by three straightest-possible lines (i.e. [geodesics](#)) can be larger or smaller than 180 degrees. The difference (the surplus or deficit), divided by the area of the triangle, is a measure for the curvature of that region of the surface.

Second possibility: the circumference of a circle. In the plane, that circumference is equal to 2 times pi times the circle's radius. On a more general surface, it can be larger or smaller. The difference, divided by the third power of the radius, leads to the same measure for the curvature as the first definition.

Simple examples for curved surfaces are the surface of a sphere (positive curvature, that is to say: sum of the angles in a triangle larger than 180 degrees, circumference of a circle smaller than 2 times pi times radius) and that of a saddle (negative curvature, that is to say: sum of the angles in a triangle smaller than 180 degrees, circumference of a circle larger than 2 times pi times radius).

Curvature cannot only be defined for surfaces, but also for higher-dimensional, more general [spaces](#) or [spacetimes](#). However, the generalized definition is substantially more complicated, and curvature is defined not by a single number, but by a set of numbers (that, together, form the “curvature tensor”). Its basic meaning, however, is the same: it measures the space's deviation from a [flat](#) space of the same dimension.

For physics, an important aspect of curvature is its connection with [gravity](#), as described in Einstein's [general theory of relativity](#). Basic information about this can be found in the spotlight text [Gravity: From weightlessness to curvature](#).

curvature singularity

A type of [spacetime singularity](#) (i.e. a boundary where [spacetime](#) ends) that is associated with infinitely strong [gravity](#) and, hence, infinitely strong [spacetime curvature](#).

Two different varieties of curvature singularity are [Ricci-](#) and [Weyl singularities](#).

More information about the different types of singularities can be found in the spotlight text [Spacetime singularities](#).

dark energy

Comparing astronomical observations with the predictions of the [big bang models](#) (which link the properties of matter and the speed of the universe's expansion), it turns out that more than 70 percent of the [density](#) of the universe is supplied by what is called *dark energy*, a type of energy that is associated with empty space itself. Ordinary matter or energy are conserved when the universe expands: If I have 10 hydrogen atoms in a certain region of space, and if that region now expands to twice its initial volume, it will still contain no more than the initial 10 hydrogen atoms, now spread over the larger volume. On the other hand, the amount of dark energy in that region of space doubles in the process, just as the volume, the "amount of space" is twice as large than it was in the beginning.

There's another crucial difference between ordinary energy and dark energy. The gravitational influence of ordinary masses and ordinary energy is attractive – it is aimed at pulling all the contents of the universe closer together. Dark energy, on the other hand, acts to accelerate the universe's expansion. In that way, it is equivalent to a certain type of what is called a [cosmological constant](#).

As yet, nobody knows how (and if) dark energy fits somewhere into our current picture of the fundamental constituents of the universe, for instance: into the [standard model of particle physics](#) or some extension of that model. This makes dark energy one of the greatest mysteries of modern physics.

dark matter

Astronomical observations of [galaxies](#) and [galaxy clusters](#) as well as comparison of observations with the predictions of the [big bang models](#) show that only about 15 percent of matter in the universe announces its presence by giving off light or other kinds of [electromagnetic radiation](#). The other 85 percent of the mass are supplied by dark matter and cosmologists have provided convincing evidence that most of that dark matter is not composed of the usual atomic constituents [protons](#) and [neutrons](#). The exact properties of these unusual matter particles are not yet known.

deflection of light, relativistic

One of the basic predictions of [general relativity](#) is that [light](#) is influenced by [gravity](#). For instance, light passing a massive body is slightly deflected. This is the basis for what is called [gravitational lensing](#).

General information about this topic can be found in the spotlight text [The gravitational deflection of light](#), while its connection with one of the fundamental principles of general relativity is examined in [The equivalence principle and the deflection of light](#).

Synonyms: relativistic deflection of light

degeneracy pressure

For a gas made up of [electrons](#), [quantum effects](#) become important. Roughly speaking, it is strictly forbidden for two electrons to be present at the same location

(this is called the [Pauli exclusion principle](#)), and if anyone attempts to concentrate electrons in a small volume of space, they will start to flit back and forth madly (cf. [Heisenberg's uncertainty principle](#)). Just like with regular gases, this flitting back and forth leads to [pressure](#), in this case to what is called degeneracy pressure.

It is this kind of electron degeneracy pressure that, for instance, stabilizes a [white dwarf](#), preventing further collapse.

Degeneracy pressure is not only possible for electrons, but for a whole class of [quantum particles](#), namely for all [fermions](#) (for example for [neutrons](#) or [protons](#)).

density

In a stricter sense synonymous with “mass density”: The average density of matter in a region of space is the total [mass](#) of all matter contained in that region, divided by the region's volume.

More generally, density can refer to other physical quantities as well. The [energy](#) density, for instance, is the total sum of energy localized in a region divided by that region's volume.

deuterium

Variant of the chemical element hydrogen where the atomic [nucleus](#) consists not of a single [proton](#), but of a proton and a [neutron](#).

In the context of [general relativity](#) it is of interest, as it is one of the species of light atomic nuclei, that formed in the early universe during [Big Bang Nucleosynthesis](#).

Deutsches Elektronensynchrotron (DESY)

Literally “German electron synchrotron” (a synchrotron being a type of particle accelerator). German research centre for [particle physics](#) and research using photons, founded in 1959 and located in Hamburg in Northern Germany. Site of the decommissioned [particle accelerator](#) HERA, among others.

[DESY website](#)

diffusion

If we put a drop of ink into clear water, then even without stirring, the ink will slowly spread throughout the water. Behind this is the motion of the ink molecules associated with the [temperature](#) of the system. The motion of each molecule is purely random, but eventually the sum of many random steps will carry a sizeable number of the molecules far away from the location where we have put the ink drop in. Processes like this where random motions lead to a spreading-out of an ensemble of molecules or other entities are called diffusion.

dimension

The number of independent directions within a set of points, alternatively: the number of [coordinates](#) needed to give each point a unique name. This is rather abstract – time for some examples:

A line is one-dimensional. There's only one direction to go on the line (the opposite direction isn't counted extra): Back-forth. A single number is sufficient to define a

point of the line. For instance, on a motorway, given the statement “the accident happened 4 kilometres from the beginning of the I95 (or M1, or whatever)” is sufficient information for the rescue workers to know exactly where to go.

[Surfaces](#) are two-dimensional, as there are two independent directions: back-forth and left-right, say. On the earth’s surface, the two coordinate numbers geographical longitude and latitude uniquely define each location.

The [space](#) that surrounds us is three-dimensional. There are three independent directions, say back-forth, left-right and up-down. In order to define a location in space, one needs to specify three numbers – for instance, two to specify where a house is located on the earth’s surface (latitude/longitude, see above) and one floor number (or, more precisely, the height above the earth’s surface).

Adding [time](#) to the three space coordinates (a must for defining an appointment – where and when?), the result is four-dimensional [spacetime](#). In order to define an event in spacetime, one needs to give four numbers: three of them determine where in space the event happens, the fourth gives the time where it happens.

According to some of the models that have been studied as candidates for a theory of [quantum gravity](#), our world should have even more space dimensions than the usual three. Some information about these [extra dimensions](#) can be found in the [spotlight topics](#) “[Extra dimensions, and how to hide them](#)”, “[Hunting for extra dimensions](#)” and [Extra dimensions and simplicity](#).

Dirac equation

Equation regulating the behaviour of [relativistic quantum particles](#) that have a spin of $1/2$, for instance [electrons](#). It was first formulated by Paul Dirac in 1928 and led directly to his successful prediction of the existence of the first species of [anti-particle](#), the positron.

Doppler effect

Effect named after the Austrian scientist Christian Doppler concerning the emission of [waves](#) by moving sources. Consider a wave-source (for instance, a device that sends out sound-waves or light-waves). Also consider two observers A and B, with observer A moving relative to the source, while observer B is at rest relative to it. When a source that moves relative to an observer emits a wave, the [frequency](#) measured by this observer is different from what a measuring instrument would record that is at rest relative to the source: If source and observer approach each other, the observer measures a higher frequency, if they move away from each other, a lower frequency.

In everyday life, the Doppler effect is readily apparent when we listen to sound waves from moving sources. If a police car or fire truck with blaring horns first races towards us, then passes us and races away, the characteristic horn sounds change dramatically in pitch, the moment the car passes us. This is because, at first, the car is moving towards us, and there is a Doppler shift towards higher pitch compared with a listener in the car. From the moment the car passes us, it becomes a source that moves away from us, with all sounds being shifted to lower pitch.

In the context of relativity, the most important Doppler effect is that for light waves. In this context, a shift towards higher [frequencies](#) is called [blueshift, one to lower frequencies redshift](#).

Synonyms: Doppler shift

$$E = mc^2$$

See [equivalence between mass and energy](#).

Synonyms: E equals m c-square

Earth

Our very own [planet](#) in the [solar system](#) – the third planet from the sun. The earth has a mass of about 6 trillion trillion (in [exponential notation](#), $6 \cdot 10^{24}$) [kilograms](#).

effective-one-body-approach

The effective-one-body (EOB)-formalism is an analytical approach describing the dynamics of two coalescing [black holes](#). It allows for the prediction of the [radiation](#) caused by such an event, which can be used in the analysis of [gravitational wave](#) signals. The EOB-approach was first introduced by Thibault Damour and Alessandra Buonanno.

Synonyms: EOB-approach, EOB-formalism

Einstein Papers Project

Project at the [California Institute of Technology](#) which is dedicated to publish a full and annotated edition of Einstein's writings and correspondence (the Collected Papers of Albert Einstein).

[Einstein Papers Project webpages](#)
[Online archive of Einstein's writings](#)

Einstein telescope

The Einstein Telescope is a proposed [third generation gravitational wave detector](#), which is supposed to deliver measurements 10 time more accurate than Advanced [Virgo](#) or Advanced [LIGO](#). Its arms are planned to be 10 kilometers long, allowing for a deeper look into space than ever before.

[Website of the Einstein telescope](#)

Einstein's equation

Einstein's equations are the cornerstone of his [general theory of relativity](#). They describe how the distortions of [spacetime](#) are connected with the properties (mass, energy, pressure...) of whatever matter is present.

Using a compact version of mathematical language, Einstein's equations, a whole system of equations, can be written in an abbreviated way so that they appear to form a single equation. That's why, sometimes, they are called "Einstein's equation" in the singular.

An elementary description of general relativity and Einstein's equations is given in the chapter [general relativity](#) of [Elementary Einstein](#).

Synonyms: Einstein's equations

Einstein@Home

Project that uses private computers to search for [gravitational waves](#) in the data of current [gravitational wave detectors](#). More information can be found on the project webpages.

> [Einstein@Home project page](#)

electric charge

The [charge](#) associated with [electromagnetism](#): a property of objects that determines the strength of the electric force with which they act on other charged objects, and the strength of electric forces by which other such objects act on them. Moving electric charges produce a [magnetic field](#), and are influenced by such fields.

electric field

The electric force is a [force](#) by which [electric charges](#) act on each other; the electric field is the associated [field](#).

Electric fields cannot be understood separate from [magnetic fields](#) – their complete description is only possible within the more general context of [electromagnetism](#).

In the simplest case, namely in situations that do not change over time, the electric force is the so-called [electrostatic force](#).

Synonyms: electric force

electrodynamics

That part of physics concerned with the study of [electromagnetism](#), in particular with [electric](#) and [magnetic fields](#) that change over time.

electromagnetic radiation

Electromagnetic influences (in the language of physics: electric and magnetic [field](#)) which, even with no electric charges present, are locked in a state of mutual excitation so that they form a [wave](#) that propagates through space.

As this wave transports [energy](#), it is, by the usual physics definition, a form of [radiation](#), called electromagnetic radiation.

Depending on frequency, there are special names for different types of electromagnetic radiation; going from lower to higher frequencies: [radio waves](#), [microwaves](#), [infrared radiation](#), [visible light](#), [ultraviolet radiation](#), [X-rays](#) and [gamma rays](#).

In the context of [quantum theory](#), it turns out that electromagnetic radiation consists of tiny energy packets, called light particles or [photons](#).

Synonyms: electromagnetic wave electromagnetic waves

electromagnetism

Electromagnetism is the totality of all phenomena associated with the presence of electric [charges](#), such as the [electric force](#), [magnetic force](#) or [electromagnetic waves](#). The basic laws of electromagnetism are [Maxwell's equations](#).

In the context of [special relativity](#) it becomes clear that electric and magnetic forces are relative – which one of them is active in a given situation depends on the observer. Say that, from the point of view of one observer, the attractive force between an electric conductor and a moving charged particle is purely electric in nature – nevertheless, for a moving observer at rest with respect to that particle, it is purely magnetic. Just as, in special relativity, it makes sense to talk of [spacetime](#) as a whole, seeing that it depends on the observer how that spacetime is then split into space and time, it makes sense to talk collectively about “the electromagnetic force”, leaving it to the different observers to split this electromagnetic force into electric or magnetic parts.

Synonyms: electromagnetic force

electron

Low-mass [elementary particle](#) with negative electric [charge](#).

The [atoms](#) that are the constituents of everyday matter each consists of a [nucleus](#) surrounded by a shell of electrons.

Synonyms: electrons

electron volt

Standard unit of [energy](#) in [particle physics](#). One electronvolt is the energy gained by an [electron](#) that is being accelerated by an electric potential difference (“electric voltage”) of 1 volt. One electronvolt, in short: 1 eV is equivalent to $1.602176 \cdot 10^{-19}$ [Joule](#) (the Joule being the energy unit of the [SI system](#) of units).

Multiples of eV that are commonly used are

kilo-electronvolt: 1 keV = 1000 eV

Mega-electronvolt: 1 MeV = 1,000,000 eV = 10^6 eV

Giga-electronvolt: 1 GeV = 1,000,000,000 eV = 10^9 eV

Tera-electronvolt: 1 TeV = 1,000,000,000,000 eV = 10^{12} eV.

Making use of the [equivalence between mass and energy](#), eV/c² is commonly used as a unit for particle masses, with c the [speed of light](#). As it is usual in particle physics to use a system of units in which light speed is equal to one, $c=1$, mass values are often simply given in eV, without explicitly mentioning the factor c².

The energy that is necessary to remove an [electron](#) from an [atom](#) is typically in the range of between a few and a few dozen eV. Typical energies of [x-ray photons](#) are in the keV range. The mass of an [electron](#) is 511 keV, that of a [proton](#) 938 MeV.

Each proton in the proton beams of the Large Hadron Collider, the particle accelerator at the CERN laboratory, is accelerated to an energy of about 7 TeV.

As the [temperature](#) is a measure of the average energy with which each component participates in a system's disordered [thermal motion](#), it can be measured in eV as well, where 1 eV corresponds to 11,604 [Kelvin](#).

Synonyms: eV keV MeV GeV

electrostatic force

An [action-at-a-distance](#)-force that acts between electrically charged bodies merely because of the fact they carry electric [charge](#).

element, chemical

A substance that cannot be decomposed into more elementary constituent substances by the methods of chemistry. From a physics point of view, to every chemical element corresponds a specific [atom](#) which is defined uniquely by the number of [protons](#) in its atomic [nucleus](#) (for example: nuclei with a single proton define the element hydrogen, two protons define helium, three Lithium, 26 iron and 92 uranium).

Synonyms: chemical element

elementary particle

Within the commonly accepted models of physics, certain particles are not built of even more fundamental sub-particles – they are themselves elementary. Examples for elementary particles are [electrons](#), [quarks](#) and [neutrinos](#), while particles such as [protons](#) or [neutrons](#) consist of sub-units and are thus not elementary. The study of elementary particles is called [particle physics](#), under which keyword some more information about the theoretical framework of elementary particles can be found.

Synonyms: elementary particles

eLISA

See [LISA](#)

ellipse

An ellipse is a [planar](#) curve. All points on the curve are located around two focal points, such that the sum of the distances to each focal point is equal for all points.

Special cases of an ellipse are a circle (in this special case, both focal points are at the same location) and a straight line joining the two focal points (in this special case, the specific distance chosen is equal to the distance between the two focal points).

As far as [gravity](#) is concerned, ellipses are of interest as the orbit of a lonely planet around a central star is an ellipse, according to [Newton's theory of gravity](#).

energy

Physical quantity with the special property that, in physical processes, energy is neither destroyed nor created, simply transformed from one form of energy to another. Some of the different forms of energy that are defined separately are [kinetic energy](#), [thermal energy](#) and the energy carried by [electromagnetic radiation](#).

Processes that transform one form of energy into another take place in all machines we use in everyday life, from the engine of a subway train (electrical energy into kinetic energy of the train) to an electric blanket (electrical energy into thermal energy).

One important consequence of [special relativity](#) is that energy and [mass](#) are completely equivalent – two different ways to define what is, on closer inspection, one and the same physical quantity. See the keyword [equivalence between mass and energy](#).

equivalence between mass and energy

Already in [special relativity](#), it turns out that (relativistic) [mass](#) and [energy](#) are really no more than two different ways of looking at one and the same physical quantity. To every form of energy, there corresponds a mass – if you heat a body up, increasing its [thermal energy](#), you automatically increase its mass. On the other hand, simply because of the mass of its constituent particles, every chunk of matter contains lots and lots of energy. In situations like the annihilation of particles in contact with their [anti-particles](#), resulting in [electromagnetic radiation](#), this matter-energy can be transformed completely into more ordinary types of energy.

The formula relating a mass to the equivalent energy is Einstein's famous

$$E=mc^2 \text{ ("E equals m-c-squared")}$$

Here E is the energy, m the corresponding [relativistic mass](#) and the constant c the [speed of light](#).

equivalence principle

One of the postulates at the basis of [general relativity](#): A freely-falling observer in a [gravitational field](#) does not feel gravity. More precisely: In a small region of space around an observer in free fall in a [gravitational field](#), the laws of physics are approximately the same as without gravitation (i.e. in [special relativity](#)) – at least for a time-limited observation period.

This is sometimes called *Einstein's equivalence principle*, which includes a more restricted version called the *weak equivalence principle*, namely that, in a gravitational field, objects which are at the same location are subject to the same gravitational [acceleration](#) – they fall at the same rate ("universality of free fall").

More information about the equivalence principle can be found in the spotlight topic [The elevator, the rocket, and gravity: the equivalence principle](#), while the path from there to Einstein's geometric gravity is traced in [Gravity: From weightlessness to curvature](#).

error (measurement)

No measurement of a physical quantity is perfectly exact – all measurements have a certain resolution or error. In particular, in measuring a physical quantity there will be a myriad of tiny disturbances (such as fluctuations in the electronic circuits used, atmospheric fluctuations in the observation of a star). Due to these disturbances, repeated measurements will sometimes give a result that is a tad too high, at other times one that is a tad too low. Such unsystematic disturbances can be described with the help of mathematical statistics. In particular, the following holds true: If the same type of measurement is repeated sufficiently often, then the results can be used to estimate both the true value of the measured quantity (the average value of the measurements) and a measure for the accuracy of the measurement (the “standard error”, also called “measurement error” or “accuracy”).

In publishing the results of measurements, it is usual to give for each result both the best approximation to the true value and an estimate of the accuracy of that approximation. Typically, a result will be written in a form such as $\gamma = 0.99983 \pm 0.00045$, the translation of which is: the best estimate for the quantity γ that results from the measurements is 0.99983, and the combined measurements have the accuracy 0.00045. Alternatively, the same result might be written as $\gamma = 0.99983(45)$, where the digits in parentheses (here 4 and 5) indicate the accuracy of the last digits of the result itself (here 8 and 3).

The error or accuracy is an estimate of the difference between the measurement result and the true value. One widely used convention (called “two sigma”) is to use an accuracy with the following meaning: Consider a measurement of some physical quantity that gives an estimate X and an accuracy Y . Then the probability that the true value of that quantity lies somewhere between $X - Y$ and $X + Y$ is 95.5 percent. With another convention (“one sigma”), the probability is only about 68 percent (but in that case, there is a probability of 95.5 percent for the true value to lie between $X - 2Y$ and $X + 2Y$). Often, estimates for systematic errors are included in the stated error as well (systematic errors do not manifest themselves in random fluctuations around the true value, but in systematic deviations – for instance, the measured value might systematically tend to be higher than the true value).

ether

In 19th century physics: Hypothetical medium in which [light](#) and other types of [electromagnetic radiation](#) propagate as [waves](#). Once the ether is postulated, certain questions arise: Does the earth move through this medium? If yes, how fast? And can this motion be detected by studying the propagation of light? Einstein’s [special theory of relativity](#), in which the value of the [speed of light](#) is independent of the motion of light source or observer (more precisely: an [inertial observer](#)), the ether turns out to be absolutely undetectable, which has led physicists to abandon the concept.

Euclidean

In a stricter sense: Euclidean geometry is the standard geometry taught in school (synonym: plane geometry). In a more general sense: Euclidean geometry is the generalization of this geometry to include three-dimensional [space](#) and even higher-dimensional spaces. The three-dimensional space we are used to in everyday life is called Euclidean space. Quite generally, spaces with three or another number of [dimensions](#) and Euclidean geometry are called [flat](#).

Synonyms: Euclidean geometry Euclidean space

European Southern Observatory

A cooperative effort of 10 member states of the European union, ESO operates a number of large-scale telescopes such as the “Very Large Telescope” (VLT) and the “New Technology Telescope” (NTT). The telescopes themselves are located in Chile; the observatory’s main office is in Garching, close to Munich in Germany.

[ESO website](#)

Synonyms: ESO

European Space Agency

The European Space Agency coordinates the space-faring activities of the European countries. It is a partner in projects such as the [Hubble space telescope](#) and the gravitational wave detector [LISA](#).

[ESA website](#)

Synonyms: ESA

event

In the context of [special](#) or [general relativity](#): Anything that is defined by a single point in time and a single point in space, i.e. something that happens at a definite time at a definite location.

Synonym: [spacetime](#) point.

event horizon

In [general relativity](#): A closed surface that is the boundary of a [black hole](#). Whatever enters through this boundary from the outside can never again leave the inside.

Near-synonym: horizon.

Event Horizon Telescope

The Event Horizon Telescope (EHT) is an array of [radio telescopes](#) that are distributed around the globe. Using Very Long Baseline Interferometry (VLBI), they detect cosmic signals. Because of their large distance to one another, they achieve a significantly better spatial resolution than a single radio telescope would. By analyzing the combined data, astronomers can observe the [event horizon](#) of a [supermassive black hole](#) such as Sagittarius A*.

exact solution

In [general relativity](#), a [solution](#) of [Einstein’s equations](#) is a model universe that follows the law of gravity given by general relativity. If the properties of that model universe can be written down explicitly, for example by expressing the geometry at each point in [spacetime](#) with the help of simple functions like fractions, square

roots, sine and cosine, the solution is called an *exact solution*. However, opinions vary somewhat as to what “writing down explicitly” means, so “exact solution”, while a useful term, is not a concept that can be defined precisely.

exponential

A quantity whose rate of growth is proportional to how large it already is is said to show exponential growth. For instance, exponential growth is a model for population growth on a planet with infinite resources: The larger the population, the more children are born and hence the larger the rate of population growth.

In the context of relativity, exponential growth is interesting for cosmology. In the hypothetical [inflationary phase](#) of our universe’s evolution the cosmos underwent exponential expansion, with the rate of expansion getting ever larger as the universe itself expanded more and more.

Synonyms: exponential growth

exponential notation

In physics, very large and very small numbers are written with the help of powers of the number ten. For large numbers, 10^n with n a non-negative integer is a 1 with n trailing zeros:

$$10^0 = 1 = \text{one}$$

$$10^1 = 10 = \text{ten}$$

$$10^2 = 100 = \text{a hundred}$$

$$10^3 = 1000 = \text{a thousand}$$

$$10^6 = 1,000,000 = \text{a million}$$

$$10^9 = 1,000,000,000 = \text{a billion}$$

$$10^{12} = 1,000,000,000,000 = \text{a trillion}$$

$$10^{15} = 1,000,000,000,000,000 = \text{a quadrillion}$$

Very small fractions – numbers that differ very little from zero – can be written with the help of 10^{-n} , with n a non-negative integer (n is, again, the number of zeros):

$$10^0 = 1 = \text{one}$$

$$10^{-1} = 0.1 = \text{one tenth}$$

$$10^{-2} = 0.01 = \text{one hundredth}$$

$$10^{-3} = 0.001 = \text{one thousandth}$$

$$10^{-6} = 0.000,001 = \text{one millionths}$$

$$10^{-9} = 0.000,000,001 = \text{one billionths}$$

$$10^{-12} = 0.000,000,000,001 = \text{one trillionth}$$

$$10^{-15} = 0.000,000,000,000,001 = \text{one quadrillionth}$$

Numbers that are not clean powers of ten can be written by factoring out the appropriate powers of ten. For instance,

$$1748 = 1.748 \cdot 1000 = 1.748 \cdot 10^3,$$

often written in what is called “scientific notation” as 1.748E3. On the other hand,

$$0.000,417,55 = 4.1755 \cdot 10^{-4} = 4.1755\text{E-}4.$$

Synonyms: scientific notation powers of ten

extra dimension

According to some of the candidate models for a theory of [quantum gravity](#), notably in [string theory](#), our world should have extra dimensions – space [dimensions](#) in addition to the three we know from everyday life.

More information about extra dimensions, possibilities of observing their effects and their use for building physical models can be found in our section [Spotlights on relativity](#), namely [Extra dimensions and how to hide them](#), [The hunt for extra dimensions](#) and [Simplicity in higher dimensions](#).

Synonyms: extra dimensions

extreme mass ratio inspiral (EMRI)

In an extreme mass-ratio inspiral (EMRI), a relatively light object orbits around a much heavier cosmic object such as a [black hole](#). EMRIs evolve very slowly and allow for detailed study of the strong [gravity](#) regime in the vicinity of the black hole using [gravitational waves](#).

Fahrenheit scale

Usual temperature scale in the US. Temperatures are given in degrees Fahrenheit (°F); the scale is defined historically, by using as its zero point the lowest temperature measured in winter 1708/1709 in Danzig (today Gdansk, Poland, Daniel Gabriel Fahrenheit’s hometown), while 100 Fahrenheit is human body temperature.

Relation with the [Celsius scale](#) widely used in Europe: X degrees Fahrenheit are $(X-32) \cdot 5/9$ degrees Celsius, Y degrees Celsius are $(Y \cdot 9/5) + 32$ degrees Fahrenheit.

Relation with the [Kelvin scale](#) widely used in science: X degrees Fahrenheit sind $(X+459.67) \cdot 5/9$ Kelvin, Y Kelvin are $(Y \cdot 9/5) - 459.67$ degrees Fahrenheit.

false colour

Only a small part of astronomical observations is concerned with visible light, in other words: with electromagnetic radiation visible to the human eye. In order to visualize observations made at invisible wave-length, such as [infra-red light](#), [radio waves](#) or [X-rays](#), the different wave-lengths are mapped to visible colours following some arbitrarily chosen scheme.

Similarly, physical quantities that are not connected with electromagnetic radiation can be mapped to colours; for instance, one can produce images of a star’s interior where different colours stand for different densities.

fermion

“Fermion” refers to [quantum particles](#) with half odd integer [spin](#), such as spin 1/2, 3/2 or 5/2.

Fermions include [elementary particles](#), for example [electrons](#) or [quarks](#), but also composite particles such as [protons](#). In more vivid terms, Fermions are particles that make up matter, while, for instance, [bosons](#) are responsible for transmitting the [elementary forces](#) between particles and belong to a different class.

Fermions are subject to the [Pauli exclusion principle](#). Two electrons can never occupy the same quantum state. This property is decisive for what we call matter: The fact that it is impossible for all electrons of an [atom](#) to occupy the lowest energy state, close to the [atomic nucleus](#), but that instead the electrons have to spread out and occupy different states, leads to the differences between atoms. This in turn is the basis for the different chemical properties of [elements](#).

field

A field describes how a physical quantity is distributed in space and time. For instance, the area where electric forces act on a [test particle](#) is subject to an [electric field](#). Or the [gravitational forces](#) which act on the mass of a test body define a [gravitational field](#). In general a field contains energy, occupies space and can change over time.

fission

See [nuclear fission](#)

flat

A [space](#) is called flat if its geometry is the direct generalization of [Euclidean geometry](#), the standard geometry taught in schools. By this definition, the simplest two-dimensional flat space is the [plane](#), and ordinary, everyday three-dimensional space is also flat, to very good approximation.

A space that isn't flat is [curved](#).

FLRW universe

See [Friedman-Lemaître-Robertson-Walker universe](#).

fluid

[State of matter](#) in which the constituent [atoms](#) and [molecules](#) are connected so loosely that the matter cannot maintain any shape without external support: If you place a fluid into a container, its shape will adapt to that of the container (in contrast with a [solid body](#), which will keep its shape). Examples of fluids are [gases](#), [liquids](#) and [plasma](#).

focus

In optics: A focus, also referred to as an image point, is the point where incoming, parallel light rays meet after traveling through a lens.

In geometry: For an [ellipse](#) all points on the curve are located around two focal points, such that the sum of the distances to each focal point is equal for all points.

force

In mechanics: Influence acting on a body, trying to [accelerate](#) it.

More generally: All influences by which [elementary](#) or other particles can interact; in this sense, force and interaction are synonymous. In the [standard model of particle physics](#), there are three elementary forces: [electromagnetism](#), the [weak \(nuclear\) force](#) and the [strong \(nuclear\) force](#), while there is no quantum description of the fourth fundamental interaction, [gravity](#).

force particle

In the framework of [relativistic quantum field theories](#) (which form the theoretical basis of the physics of [elementary particles](#), the forces by which matter particles interact are transmitted by so-called carrier particles travelling back and forth between them. For instance, the [electric force](#) between two [electrons](#) would come about through the exchange of [photons](#), the carrier particles of the [electromagnetic interaction](#). Carrier particles always have integer [spin](#), such as spin 1 or 2 (which means they belong to the class of particles called [bosons](#)). Synonym: carrier particles.

fourth test of general relativity

Another name for measurements of the [Shapiro time delay](#) as an addition to the three [classical tests](#) of [general relativity](#).

Synonyms: fourth test of general relativity

frame of reference

See [reference frame](#)

frame-dragging

In [Newtonian gravity](#), the [gravitational field](#) of a [mass](#) is independent of whether or not that mass rotates. In [general relativity](#), a mass's rotation influences the motion of objects in its neighbourhood. Put simply, the rotating mass "drags along" spacetime in the vicinity.

This is known as Lense-Thirring effect or frame-dragging. Sometimes, frame-dragging is also used in a more general sense that includes additional general-relativistic effects associated with the movement of sources of gravity. There is an analogy between gravity and [electromagnetism](#) in which ordinary gravity corresponds to the [electrostatic force](#), and the field components responsible for frame-dragging to *magnetism*. For this reasons, these effects also go by the name of gravitomagnetism.

Synonyms: Lense-Thirring effect, Gravitomagnetism

free

In the context of [relativity theory](#), a particle (object, [observer](#)...) that is not acted upon by any force except [gravity](#) is said to be free or, a bit more specific, to be in free fall. Free [test particles](#) play an important role in understanding the structure of [general relativity](#).

Synonyms: free fall, free particle

frequency

Measure for the rapidity of an oscillation, defined as the inverse of the period of oscillation: A process that, in oscillating, repeats itself after 0.1 seconds has the frequency $1/(0.1 \text{ seconds}) = 10 \text{ Hz}$. (The unit Hertz, abbreviated as Hz, is defined as $1 \text{ Hz} = 1/\text{second}$.)

For a simple wave, the frequency is given by the number of maxima going by a stationary observer in a second. Ten maxima going by per second correspond to a frequency of 10 Hz.

Friedmann-Lemaître-Robertson-Walker universe

The simplest assumptions one can make about a universe are that it is homogeneous and isotropic. Homogeneity means that the properties of [matter](#) and of the [geometry](#) of [spacetime](#) are the same at every point in [space](#). Isotropy means that all spatial directions are on the same footing, and that to a hypothetical observer, such a universe looks exactly the same, in whatever direction he or she might be looking. These assumptions are quite restrictive; in fact, it is possible to write down an expression characterizing the spacetime geometry of all homogeneous and isotropic [solutions](#) of [Einstein's equations](#). The result is a family of spacetimes known as *Friedmann-Lemaître-Robertson-Walker universes*. Typically, these universes are either in a state of expansion or a state of collapse. The best-known example is the expanding universe described by [big bang](#) cosmology.

Sometimes, these model universes are also referred to as Friedmann-Lemaître universes, Robertson-Walker universes or Friedmann-Robertson-Walker universes.

fusion

See [nuclear fusion](#)

galaxy

[Stars](#) are rarely found alone – usually, they congregate in conglomerates of millions, billions or even more stars called galaxies. A case in point is our sun, part of a galaxy we call the [milky way](#).

The life of a young galaxy can be very turbulent. Examples of such young, [active galactic nuclei](#) are [radio galaxies](#) and [quasars](#).

galaxy cluster

Galaxies are not solitary objects – usually, they cluster together. Our own galaxy for instance, the Milky Way, is part of a small cluster called the [local group](#) of galaxies. The next-closest large galaxy cluster is the [Virgo cluster](#).

gamma rays

The most highly energetic variety of [electromagnetic radiation](#), with over a quintillion oscillations per second, corresponding to [wave-lengths](#) of less than a hundredth billionth of a metre.

Synonyms: gamma radiation

gamma-ray burst

Astronomical events visible as extremely strong flashes of [gamma rays](#). Their origin is still unclear; In the context of [general relativity](#) they are interesting because they are thought to signal the mergers of [neutron stars](#) and/or [black holes](#). Shortly after the observation of the [gravitational wave](#) GW170817 a gamma ray burst was detected and thus underpins the hypothesis.

Synonyms: gamma burst

gas

In a strict sense: A [state of matter](#) in which the [atoms](#) and/or [molecules](#) wildly careen and collide, without being bound to each other. This movement leads to an inner [pressure](#), while the average [kinetic energy](#) of the moving particles is a measure for the [temperature](#) of the gas.

Compare the other [states of matter](#): [solid state](#), [liquid](#), [plasma](#).

In a broader sense, gas is also used to denote other mixtures of freely careening particles, for instance in the case of the [electron](#) gas whose pressure stabilizes a [white dwarf](#) against further collapse.

general theory of relativity

Albert Einstein's theory of [gravity](#); a generalization of his [special theory of relativity](#).

For information about the concepts and applications of this theory, we recommend the chapter [general relativity](#) of our introductory section [Elementary Einstein](#). Further information about many different aspects of general relativity and its applications can be found in our section [Spotlights on relativity](#).

Synonyms: general relativity

GEO600

British-German gravitational wave detector located in Ruthe (close to Hannover, Germany). GEO600 is an [interferometric gravitational wave detector](#) with an arm-

length of 600 metres.

[GEO600 website](#)

geodesic

A straightest-possible line in a [surface](#) or a more general [space](#). In the [plane](#), the geodesics are [straight lines](#), on the surface of a sphere they are [great circles](#).

geodetic precession

In [classical mechanics](#), the rotation axis of a gyroscope on which no external forces are acting will remain constant – a useful property that has found applications in navigation. However, in the presence of [spacetime curvature](#), this is no longer true – the axis direction of a gyroscope in free fall will change over time; an effect predicted by Einstein's [general theory of relativity](#).

geometry

That part of mathematics concerning itself with [surfaces](#) or more general [spaces](#) as well as objects defined on such spaces, such as [points](#) or [lines](#) as well as the objects constructable from points and lines, such as [triangles](#).

GeV

See [electron volt](#)

Synonyms: Giga-electronvolt

global positioning system

A system of satellites and mobile receivers that makes it possible to determine each receiver's position with high accuracy. Used by pilots, truckers, car drivers and by many smartphone users world-wide, it is an industrial application of Einstein's theories of [special](#) and [general relativity](#): Without taking into account the effects predicted by these theories for moving clocks in a gravitational field, there would be errors of roughly 10 kilometres per day of operation in the determination of positions on earth.

Synonyms: GPS

gluons

The [carrier particles](#) of the [strong nuclear force](#). They are responsible for binding (glueing) [quarks](#) together into compound particles like [protons](#) or [neutrons](#).

gold

Chemical [element](#) with the symbol Au; each gold [nucleus](#) contains 79 [protons](#).

Gold nuclei, stripped of their [electrons](#), are among the types of heavy [ions](#) which are brought into collision in [particle accelerators](#) such as the [Relativistic Heavy Ion Collider](#) in order to recreate the state of matter in the early universe shortly after the [big bang](#).

Gowdy spacetimes

Gowdy spacetimes (or universes) are simple expanding model universes. In contrast with the better known [Friedmann-Lemaître-Robertson-Walker](#) universes (the basis for the [big bang models](#)), Gowdy universes are not homogeneous. Instead, they are filled with a regular pattern of [gravitational waves](#). A Gowdy T3 universe is the simplest kind of Gowdy universe, in which space has the shape of a three-dimensional [torus](#).

More information about Gowdy universes can be found in the spotlight text [Of gravitational waves and spherical chickens](#).

Synonyms: Gowdy universes Gowdy T3

gravitation

In [classical](#) physics: An [action-at-a-distance](#) force by which all bodies that possess [mass](#) attract each other (see [Newtonian theory of gravity](#)), synonym: gravitational force.

In Einstein's [general theory of relativity](#): The fact that matter that possesses [mass](#), [energy](#), [pressure](#) or similar properties distorts [spacetime](#), and that this distortion in turn influences whatever matter might be present.

An introduction to the basic ideas of general relativity is provided by the section [General relativity](#) of [Elementary Einstein](#). More information about the nature of gravity in general relativity can be found in the spotlight text [Gravity: From weightlessness to curvature](#).

Synonyms: gravity

gravitational constant

Constant of nature; the fundamental [Newton's law of gravity](#) and thus a measure for the natural strength of gravity. Analogously, in [Einstein's equations](#) in the [general theory of relativity](#), it occurs as the proportionality factor determining how strongly [mass](#), [energy](#) and similar properties of matter distort space and time. In formulae, it is usually written as G. The best current value for G is

$$G = 6.674\,30(15) \cdot 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}.$$

Compared with other fundamental constants, G is known only to a comparatively low accuracy.

gravitational field

The totality of all gravitational influences that one or more massive objects can exert on bodies in their vicinity.

More precisely: At every location in space, the gravitational field is defined as the [acceleration](#) that a small [test particle](#) present at that location would feel due to the gravitational forces of the masses around it.

gravitational lens

In Einstein's [general relativity](#), gravity necessarily acts not only on material bodies, but also on [light](#) – light passing a massive body is deflected. This deflection can be so strong that light of one and the same cosmic object reaches an observer along multiple paths – corresponding to the observer seeing multiple images of that object in the sky. Masses that, in this sense, act like very special optical lenses are called gravitational lenses.

More information can be found in the spotlight text [A brief history of gravitational lenses](#).

Synonyms: gravitational lensing

gravitational redshift

According to [general relativity](#), [light](#) flying away from a massive body (or other source of [gravity](#)) experience a [redshift](#) – its [frequency](#) decreases and the light becomes less energetic. On the other hand, light flying towards a massive body gets blueshifted – its frequency and [energy](#) increase.

gravitational wave astronomy

Discipline of astronomy which aims at using [gravitational waves](#) to gain information about cosmic objects or the cosmos as a whole – for instance about what's happening in the core region of a [supernova](#), about [neutron star](#) or about the heated past of our universe.

Since 2015, scientists have been [detecting gravitational waves](#) using highly sophisticated [gravitational wave detectors](#), so that the time of gravitational astronomy has now begun for real.

gravitational wave detector

Currently, scientists world wide are attempting the direct measurement of [gravitational waves](#) reaching us from the depths of space. They are mainly using two types of detectors: [interferometric detectors](#) like [GEO600](#) and the [LIGO detectors](#), and [resonant detectors](#).

For more informations about gravitational waves, please consult the chapter [Gravitational waves](#) of [Elementary Einstein](#).

gravitational waves

Distortions of space geometry that propagate through space with the speed of light, analogous to ripples on the surface of a pond propagating as water waves.

For more informations about gravitational waves, please consult the chapter [Gravitational waves](#) of [Elementary Einstein](#).

Selected aspects of gravitational wave physics are described in the category [Gravitational waves](#) of our [Spotlights on relativity](#).

Gravitomagnetism

See [frame-dragging](#).

Synonyms: frame-dragging Lense-Thirring effect

graviton

Hypothetical [carrier particle](#) in a [quantum theory](#) of [gravity](#). However, as of yet physicists have but a rough idea of how a complete theory of [quantum gravity](#) will look like.

gravity

See [gravitation](#)

GRAVITY

GRAVITY is part of the Very Large Telescope Interferometer (VLTI) at the [European Southern Observatory](#) in Chile. The instrument operates in the near infrared and combines the light of four telescopes, whereby it achieves the spatial resolution of an equivalent telescope with 130 meters in diameter. Due to its high sensitivity and accuracy, it allows researchers to render very small and faint cosmic objects visible. GRAVITY also measures distances very accurately and is therefore very suitable for the measurement of movements. For example, with the help of GRAVITY, the motion of a [star](#) around the [black hole](#) at the centre of our galaxy could be observed, confirming the predictions of [general relativity](#).

great circle

Circle on the surface of a sphere whose center coincides with that of the sphere itself. On the globe, the equator is a great circle, while every meridian corresponds to half of a great circle.

If you want to move on a spherical surface in the straightest possible way, choose a path along a great circle – in the language of mathematics this is equivalent to saying: great circles are [geodesics](#) of a spherical surface.

Hawking radiation

[Thermal radiation](#) emitted by [black holes](#) due to [quantum effects](#). First calculated by the British physicist Stephen Hawking in the 1970s. The characteristic [temperature](#) of the radiation, which depends on the mass and spin of the black hole, is called [Hawking temperature](#).

Hawking temperature

Characteristic [temperature](#) of the [Hawking radiation](#) of a [black hole](#). For simple, spherically symmetric black holes, it is

$TH = 6 \cdot 10^{-8}$ ([solar mass](#)/mass of the black hole) [Kelvin](#). [Problems reading expressions such as 10^{-8} ? See [exponential notation](#).]

Heisenberg's uncertainty principle

Is a fundamental law of [quantum theory](#), which defines the limit of precision with which two complementary physical quantities can be determined. If one of the quantities is measured with high precision, the corresponding other quantity can necessarily only be determined vaguely. In other words, it is impossible to measure simultaneously both complementary quantities with greater precision than the limit defined by the Heisenberg's uncertainty principle.

An example for such complementary quantities are the location and the momentum of a [quantum particle](#): Very precise determination of the location make precise statements about its momentum impossible and vice versa.

helium

After [hydrogen](#), the second lightest chemical [element](#). Its atomic nucleus consists of two [protons](#) and, ordinarily, two [neutrons](#) ("helium-4"); such helium [nuclei](#) are also called alpha particles. Another variety of helium, helium-3, has only one neutron in its nucleus.

In the context of [general relativity](#), both helium-3 and helium-4 are of interest as two species of light atomic nuclei that formed in the early universe during [Big Bang Nucleosynthesis](#).

Hertz

Unit of [frequency](#), abbreviation: Hz. One Hertz corresponds to one oscillation per [second](#).

horizon

In [general relativity](#): A closed surface that is the boundary of a [black hole](#). Whatever enters through this boundary from the outside can never again leave the inside.

Synonym: event horizon.

Synonyms: event horizon

Hubble constant

In an expanding universe such as that of the [big bang models](#), every observer will find: The apparent velocity with which the galaxies around him recede is proportional to their distance; the more distant a galaxy, the more its distance increases in a given time. This relation was first found by the astronomer Edwin Hubble in the 1920s from observations of far-away galaxies; it is hence called Hubble relation or Hubble's law, and the constant of proportionality between speed and distance is the [Hubble constant](#).

A visualisation of the Hubble relation can be found on the page [The expanding universe](#) in the chapter on [Cosmology](#) of [Elementary Einstein](#).

The Hubble relation only holds for all galaxies in an idealized universe whose expansion neither accelerates nor slows down. In more realistic universes, it is true in good approximation only for galaxies that are not too far away.

Synonyms: Hubble effect Hubble relation Hubble's law

Hubble relation

See [Hubble constant](#)

Hubble space telescope

Cooperative project of [NASA](#) and [ESA](#): Space telescope that was put into orbit in 1990. Orbiting 600 kilometres above the earth, it leaves behind the densest parts of the earth's atmosphere, allowing an unrivalled, undisturbed view into space.

[Outreach website for the Hubble Space Telescope](#)

[Website of the Space Telescope Science Institute](#)

hydrogen

The lightest (and, in our universe, the most abundant) chemical [element](#). The atomic [nucleus](#) of an ordinary hydrogen atom is a single [proton](#). If the atomic nucleus contains an additional [neutron](#), the atom is called *heavy hydrogen* or *deuterium*.

imaginary time

Certain quantum calculations (notably the calculation of [path integrals](#) as a way to find quantum mechanical probabilities) involve an algebraic manipulation of the following kind: Wherever the time coordinate t occurs, it is replaced by $i \cdot t$, where i is the “imaginary unit”, a number defined to have the remarkable property $i^2 = i \cdot i = -1$. At the end of the calculation, the substitution is reversed. The combination $T = i \cdot t$ is called imaginary time.

Most such calculations occur in [particle physics](#), in the framework of [special relativity](#), where there are rigorous mathematical proofs showing how the use of imaginary time leads to correct results.

Imaginary time has also been employed in some candidate theories for a theory of [quantum gravity](#), notably in certain types of [quantum cosmology](#). This, however, involves the flexible time of [general relativity](#), and both the details of the imaginary time recipe and the more general question whether or not imaginary time can usefully be employed in this context in the first place are still unresolved, and the object of current research.

Some more information about path integrals and the role of imaginary time can be found in the spotlight text [The sum over all possibilities](#), while imaginary time in quantum cosmology is briefly discussed in [Searching for the quantum beginning of the universe](#).

inertia

Basic law of [classical mechanics](#) and [special relativity](#): bodies on which no external forces act move with constant speed on straight paths. In the geometrical language

of special relativity, this can be reformulated as: bodies on which no external forces act move on [straight line](#) in [spacetime](#).

Strictly speaking, though, this law is only true in specific [reference frames](#). This gives rise to a further, more precise reformulation: It is always possible to find a reference frame on which bodies on which no external forces act move with constant speed along straight paths. Such reference frames are called [inertial frames](#).

In [general relativity](#), the law of inertia holds in a somewhat modified form: there, bodies on which no external, non-gravitational forces act don't move on straight lines through spacetime, but on [geodesics](#).

Synonyms: law of

inertial forces

In [classical mechanics](#) or [special relativity](#): Whenever an [observer](#) who is not an [inertial observer](#) wants to explain the movements of bodies using the law “force equals mass times acceleration”, that observer has to assume the existence of additional forces; these are called inertial forces. For ordinary forces like the [electric force](#), the [magnetic](#) or the [gravitational](#) force, one can always state which bodies are acting on which other bodies; inertial forces, in contrast, appear to act on bodies “from nowhere”.

A famous example for an inertial force is the [centrifugal force](#) – an observer riding a merry-go-round needs to introduce that force to explain why he and all other riders are pulled away from the axis of rotation.

inertial observer

An inertial reference frame is a [reference frame](#) in which the first law of [classical mechanics](#) holds: A body on which no external forces act either remains at rest or moves with constant speed along a straight path. An inertial observer is an observer that is at rest with respect to an inertial reference frame. In the context of relativity, an inertial reference frame is one that drifts in gravity-free space without undergoing rotation or being accelerated.

Inertial reference frames play a central role in [special relativity](#): the basic postulates of that theory are the [relativity principle](#) (which holds that the laws of physics are the same in all inertial reference frames – no such frame is special, in this sense) and the postulate that the speed of light has the same value for every inertial observer.

In [general relativity](#), there are no real inertial observers, however, by what's called the [equivalence principle](#), the laws of physics for an observer that is in free fall and performs his measurements only in his direct neighbourhood (and only over a limited period of time), the laws of physics are approximately the same as for an inertial observer.

Synonyms: inertial reference frame

inflation

Hypothetical phase in the earliest universe during which the cosmos underwent [exponentially growing](#) expansion.

Synonyms: inflationary phase

infrared

[Electromagnetic radiation](#) in the [frequency](#) region between a hundred billion and a trillion oscillations per second, corresponding to [wave-lengths](#) between 0.8 [micrometres](#) and 1 [millimetre](#). The [thermal radiation](#) associated with everyday temperatures is infrared radiation.

Synonyms: infrared light IR

inspiral

The inspiral is one phase in the life cycle of binary [black holes](#). The objects lose [energy](#) due to the emission of [gravitational waves](#), which causes their orbits to gradually shrink. They spiral towards each other with increasing [speed](#), until they collide and [merge](#).

INTEGRAL

Satellite launched in 2002 as a space-borne [gamma ray](#) observatory by the [European Space Agency](#).

[Public outreach pages of the INTEGRAL mission](#)

interaction

Interactions are all the different ways in which [elementary](#) or compound particles can influence each other. In elementary particle physics, “interaction” and “force” are used synonymously.

In the [standard model of elementary particles](#), there are three fundamental interactions: [electromagnetism](#), the [strong nuclear force](#) and the [weak nuclear force](#). For another interaction, [gravity](#), there is no quantum description yet.

interference

When [waves](#) meet and are superimposed, they can amplify or dampen each other. These superposition effects are called interference effects: Wherever a wave-crest meets a wave-crest, a higher wave-crest results (constructive interference); when a wave-crest meets a trough, there can be a complete cancellation between the two (destructive interference).

Interference can happen among [electromagnetic waves](#) (such as light), but also among water waves and among sound waves.

interferometric detector

[Gravitational wave detector](#) that utilizes [interference](#) between light waves to detect minute changes in distance effected by a passing [gravitational wave](#).

Some of the physics behind interferometric detectors can be found in our spotlight topic [Catching the wave with light](#).

Exmples for interferometric detectors are [GEO600](#) and the [LIGO detectors](#).

Synonyms: interferometric gravitational wave detector

intergalactic medium

A thin [gas](#) that fills some parts of the empty regions between [galaxies](#). The distribution is non-uniform; filaments of intergalactic medium are separated by voids with much lower [density](#). The main ingredient of the intergalactic medium is [ionized hydrogen](#), in other words: a [plasma](#) consisting of an equal number of hydrogen [nuclei](#) ([protons](#)) and [electrons](#). The average density of the intergalactic medium is estimated to be between ten and a hundred hydrogen atoms per cubic metre.

International Space Station

Space station in an earth orbit, constructed as a cooperative project of 16 different nations. In the context of relativity, its main reason is to be an example for a laboratory in free fall in the earth's gravitational field.

[ISS pages \(NASA\)](#)

Synonyms: ISS

International System of Units

See [SI, International System of Units](#)

Synonyms: SI

ion

Usually, an [atom](#) possesses as many electrically positive [protons](#) in its [nucleus](#) as it has electrically negative [electrons](#) in its shell, rendering it, overall, electrically neutral. Atoms that have more or fewer electrons and are thus, as a whole, electrically [charged](#), are called ions. If you start with an atom that is electrically neutral and make it into an ion by removing or adding electrons, you have ionized that atom.

Synonyms: ionize

isotope

Different species of atomic [nuclei](#) which contain the same number of [protons](#) (and thus represent the same chemical [element](#)), but different numbers of [neutrons](#), are called isotopes.

Thus, [helium](#)-4 (two protons, two neutrons) and helium-3 (two protons, one neutron) are isotopes, but helium-3 and [tritium](#) (one proton, two neutrons) are not. The word can also be used of a specific species of nucleus and a generic chemical element, as in “helium-3 is an isotope of helium”.

jets

In the context of astronomy: strongly focussed, highly energetic particle streams like those emitted by certain [active galactic nuclei](#). Jets can become visible when they deposit their energy in a huge gaseous region, making them shine out bright (so-called radio bubbles).

Joule

The unit of [energy](#) in the International System of units ([SI](#)). One Joule is equal to one [kilogram](#) times square [metre](#) over square [second](#), in short: $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$. It is equal to the [kinetic energy](#) gained by an object of the mass one kilogram that has been accelerated with an [acceleration](#) of one metre per square second over a distance of one metre.

KAGRA

The Kamioka Gravitational Wave Detector (KAGRA) is an underground Japanese [gravitational wave detector](#), the length of its two arms being 3 km. KAGRA uses cryogenic technology, which means that the interferometer’s mirrors are cooled down to 20 Kelvin in order to reduce thermal noise. On top of that, the underground construction is expected to reduce seismic noise. The detector has started observation in 2020.

Karlstad University

University (enrolment ca. 10,000) in central Sweden. Research areas include a number of relativistic topics, from [string theory](#) to the mathematics of [general relativity](#).

[Karlstad University website](#)

Kelvin scale

The temperature scale used in physics, synonym: absolute temperature.

The zero point of the Kelvin scale is at [absolute zero](#); a temperature difference of one Kelvin (abbreviated 1 K and, rarely, also called one “degree Kelvin”) is the same as a temperature difference of one degree [Celsius](#), as both scales differ only by their choice of zero point: X degrees Celsius are (X plus 273.15) Kelvin, Y Kelvin are (Y minus 273,15) degrees Celsius.

Relation to the [Fahrenheit scale](#): X degrees Fahrenheit are $(X+459,67)*5/9$ Kelvin, Y Kelvin are $(Y*9/5)-459,67$ degrees Fahrenheit.

Kepler's laws

Basic laws governing the orbital motions of [planets](#) around the [sun](#). First law: Each planetary orbit is an [ellipse](#), with the sun in one of its [focus points](#). Second law: If you connect the planet and the sun by an imaginary line then, in equal time intervals, the line will sweep over equally large areas, independent on where the planet is on its orbit. Third law: dividing the square of a planet's orbital period by the third power of it's average distance from the sun gives the same value for all planets in the solar system; written as a formula: $\text{period}^2 / (\text{average distance to the sun})^3 = \text{const.}$

Kepler's laws follow directly from the laws of [classical mechanics](#) and [Newton's law of gravity](#). However, they are only valid approximately – the gravitational pull of the planets on each other, as well as the fact that, ultimately, gravity is governed not by Newton's laws, but by [general relativity](#) (see relativistic [perihelion shift](#)) lead to small deviations from perfectly elliptic orbits.

Synonyms: Kepler's laws of planetary motion

Kerr black hole

The simplest kind of a rotating black hole: a [model universe](#) containing a single rotating black hole and nothing else. This solution to [Einstein's equations](#) was found by Roy Kerr in 1963.

Synonyms: Kerr solution

Kerr-Newman black hole

The simplest kind of a rotating, [electrically charged](#) black hole: a model universe containing a single rotating, charged black hole and nothing else. This solution to [Einstein's equations](#) was found independently by Roy Kerr and Ted Newman in 1963.

The Kerr-Newman solution is more general and includes the [Kerr solution](#) if the charge of the black hole is zero.

Synonyms: Kerr-Newman solution

keV

See [electron volt](#)

Synonyms: kilo-electronvolt

kilogram

In the international system of units ([SI](#)), the unit of [mass](#); until May 2019 defined by a reference mass that is kept in Paris, France. From then on by definition of the Planck constant and of the Avogadro number that has an exact experimental value used for the definition of the Mol.

kinetic energy

A type of [energy](#) that has to be ascribed to an object simply because that object moves relative to the [reference frame](#). In [classical](#), pre-Einstein physics, the amount of energy is given by one half, times an object's [mass](#), times the square of its [speed](#).

Klein-Gordon equation

Equation regulating the behaviour of [relativistic quantum particles](#) with [spin](#) 0.

Lamb shift

Typically, [electromagnetic radiation](#) is emitted by [atoms](#) only at very specific frequencies that depend on the type of atom. For a number of these characteristic frequencies, the [relativistic quantum theory](#) of [electromagnetism](#) (called [quantum electrodynamics](#)) predicts a slight shift, as compared with earlier theories. This is the Lamb shift. Experiments have confirmed the prediction.

Large Hadron Collider

The Large Hadron Collider is a [particle accelerator](#) in the research centre [CERN](#). From the point of view of relativity theory, it has several points of interest: First of all, it accelerates [protons](#) to higher energies than ever, allowing new tests of the relativistic [quantum field theories](#) that are at the core of modern [particle physics](#). Secondly, at such high energies, there should be first traces of an as-yet unproven symmetry of nature called [supersymmetry](#), which plays an important role in [string theory](#), one of the candidates for a theory of [quantum gravity](#) (the [quantum theory](#) version of Einstein's [general relativity](#)). Finally, the high energies are interesting because they give information about the very early high temperature universe, and about the physics that should be included in the [big bang](#) models of relativistic cosmology.

[CERN website](#)

Synonyms: LHC

laser

Abbreviation for "Light Amplification by Stimulated Emission of Radiation". Technique for the production of very concentrated, strong light with a fixed [frequency](#), which propagates as a very simple [electromagnetic wave](#) in which wave crests and wave troughs are in perfect step ("coherent light").

Laser Interferometer Gravitational Wave Observatory

See [LIGO](#)

Laser Interferometer Space Antenna

See [LISA](#)

lead

Chemical [element](#) with the symbol Pb; each lead [nucleus](#) contains 82 [protons](#).

Lead nuclei, stripped of their [electrons](#), are among the types of heavy [ions](#) which are brought into collision in [particle accelerators](#) such as the [Relativistic Heavy Ion Collider](#) in order to recreate the state of matter in the early universe shortly after the [big bang](#).

length contraction

Effect of [special relativity theory](#): An observer (more precisely: an [inertial observer](#)) measures a shorter length for a moving object than for an identical copy of that object resting beside him (here, length refers to extension in the direction of movement – extension in orthogonal directions remains the same).

Lense-Thirring effect

See [frame-dragging](#).

Synonyms: frame-dragging Gravitomagnetism

light

Light in the strict sense of the word is [electromagnetic radiation](#) the human eye can detect, with wave-lengths between 400 and 700 [nanometres](#). In relativity theory and in astronomy, the word is often used in a more general sense, encompassing all kinds of electromagnetic radiation. For instance, astronomers might talk about “infrared light” or “gamma light”; in this context, light in the stricter sense is referred to as “visible light”.

Within [classical](#) physics, the properties of light are governed by [Maxwell's equations](#); in [quantum physics](#), it turns out that light is a stream of energy packets called light quanta or [photons](#).

In the context of [relativistic physics](#), light is of great interest, and for a number of reasons. First of all, the [speed of light](#) plays a central role in both [special](#) and [general relativity](#). Also, there are a number of interesting effects in general relativity which are associated with the propagation of light, namely [deflection](#), the [Shapiro effect](#) and the [gravitational redshift](#).

light elements

According to the [big bang models](#), the early universe underwent a brief period of primordial [nucleosynthesis](#) between a few seconds and a few minutes [cosmic time](#), during which nuclei of light elements such as heavy hydrogen, helium and lithium formed.

A brief account of this *Big Bang Nucleosynthesis* can be found in the spotlight text [Big Bang Nucleosynthesis](#), while [Equilibrium and change](#) provides more information about the physical processes involved and [Elements of the past](#) describes how the predictions of Big Bang Nucleosynthesis can be tested against astronomical observation.

Synonyms: origin of light elements

light speed

See [speed of light](#).

light-cone

In [special](#) as well as in [general relativity](#), the [speed of light](#) sets the upper limit for the transmission of influences and signals. Thus, studying the propagation of light, one can find out for an event A which other events can influence A, which can be influenced by A, and where any influence is impossible (because such influence would have to travel faster than light – cf. [causal structure](#)). Graphically, the boundary between the two sets of events where influence is possible or impossible has the form of a double cone (for a sketch, see the page [Spacetime](#) in the chapter [Special relativity](#) of [Elementary Einstein](#)). It is formed by all [world-lines](#) of hypothetical light signals that would be emitted at the event A or, coming from an arbitrary direction, would be absorbed there.

Light-second

Units of distance: A light-year is the distance a light signal travels during one year of flight-time, a light-second the distance it travels in a second.

The following table translates these units into the more familiar kilometres or seconds: 1 light-second = 300000 km = 186000 mi. 1 light-minute = 18 millionen km = 11 million mi. 1 light-hour = 1.1 billion km = 670 million mi. 1 light-day = 25 billion km = 16 billion mi. 1 [light-year](#) = 9.5 trillion km = 6 trillion mi.

Synonyms: light-minute light-hour light-day light-year

light-wave

Light is a mixture of elementary [waves](#), each a regular succession of maxima and minima of the [electromagnetic field](#). The same is true for all other sorts of [electromagnetic radiation](#).

light-year

See [light-second etc.](#), above

Synonyms: light-years

LIGO

Laser Interferometer Gravitational-Wave Observatory : A detector project for the measurement of [gravitational waves](#) in the United States, which was upgraded to the [Advanced LIGO](#) between 2010 and 2015 .The detector includes two [interferometric gravitational wave detectors](#), with an arm-length of four kilometers each. One of them is located in Hanford, Washington, the other one is located in Livingston, Louisiana. [The first ever direct measurement of gravitational waves](#) was made at LIGO.

> [LIGO website](#)

Synonyms: Laser Interferometer Gravitational-Wave Observatory

line

Geometric object with a single [dimension](#). A line can either be an independent one-dimensional [space](#) (in the abstract mathematical space where a space does not need to have three dimensions), or it can be embedded into a more general [space](#), like a line drawn onto a piece of paper (i.e. a surface).

liquid

[State of matter](#) in which the constituent [atoms](#) and [molecules](#) are bound to each other, but so loosely that the matter cannot maintain any shape without external support: If you place a liquid into a container, its shape will adapt to that of the container (in contrast with a [solid body](#), which will keep its shape, and in common with other [fluids](#) such as [gas](#) or [plasma](#)).

Compare the other [states of matter](#): [solid state](#), [gas](#), [plasma](#).

LISA

The concept of LISA (Laser Interferometer Space Antenna), a space-based [gravitational wave detector](#), has been studied jointly by the European and US space agencies [ESA](#) and [NASA](#) for 20 years. The planned interferometric gravitational wave detector should be composed of three satellites, positioned to form a triangle, with each side 2.5 million kilometers long. While called eLISA for some time, the project named LISA is expected to launch in 2034.

> [LISA website](#)

Synonyms: Laser Interferometer Space Antenna

LISA Pathfinder

LISA Pathfinder was a test mission of the European Space Agency ESA for the [LISA mission](#). LISA Pathfinder demonstrated the functionality of crucial LISA technologies with which the first gravitational wave observatory in space will observe low-frequency gravitational waves.

lithium

Chemical [element](#) whose atomic [nucleus](#) contains three [protons](#).

In the context of [general relativity](#), it is of interest as one of the light elements that formed in the early universe during [Big Bang Nucleosynthesis](#).

local group

The [galaxy cluster](#) of which our very own [galaxy](#), the [Milky Way](#), is a member. Cosmically speaking, the local group is rather puny – its only members apart from the Milky Way are the Andromeda galaxy, the galaxy M33 and a number of dwarf galaxies (such as the Magellanic clouds).

Lockman hole

A region in the sky, located in the constellation Ursa Major (better known as the Big Dipper), and covering an area about 75 times that of the full moon. Pointing their telescopes at this region, astronomers will encounter only negligible amounts of the [hydrogen](#) gas that fills significant portions of our galaxy. These are ideal conditions for obtaining a clear and unobstructed view of objects in deep space, far beyond our own galaxy. Not surprisingly, the Lockman hole is one of the most intensively studied regions in the night sky.

logarithmic

In graphic representations of physical data, a way of plotting values according to their common logarithm. In mathematics, the logarithm is the inverse function to exponentiation. That means the logarithm of a given number x is the exponent to which another fixed number, the base b , must be raised, to produce that number x . If the base is 10 then the logarithm of 10 is 1, the logarithm of 100 is 2, and so on.

In ordinary (“linear”) plots, the distance between the values 1 and 2 is the same as between the values 2 and 3. In a logarithmic plot, the distance between the values 1 and 10 is the same as the distance between 10 and 100 and the distance between 100 and 1000.

Synonyms: logarithmic scale

lookback time

The time that [light](#) from a distance body needs to reach us here on earth. Lookback time, defined using the [cosmic time coordinate](#), is one way of defining distances in an expanding universe.

loop quantum gravity

Candidate for a theory of [quantum gravity](#). Loop quantum gravity is an attempt to apply the concepts and laws of [quantum theory](#) directly to the geometry that is at the heart of [general relativity](#).

A brief description can be found on the page [Loop quantum gravity](#) in the chapter [Relativity and the quantum](#) of [Elementary Einstein](#).

Lorentz transformation

Central set of formulae of [special relativity](#): Formulae that define how to go back and forth between two [inertial reference frames](#) that are in relative motion; more precisely: if an event is defined in terms of the space- and time coordinates of one of the observers, one can calculate which [coordinates](#) the other observer would assign to that same event.

magnetic field

The magnetic force is a force by which electric currents (i.e. moving electric [charges](#)) act on each other; the magnetic field is the associated [field](#). All phenomena related to the magnetic force or magnetic field are subsumed under the heading of *magnetism*. Magnetic fields cannot be understood separate from [electric fields](#) –

their complete description is possible only within the more general context of [electromagnetism](#).

Synonyms: magnetic force magnetism

mass

In [classical](#) physics, mass plays a triple role. First of all, it is a measure for how easy it is to influence the motion of a body. Imagine that you're drifting in empty space. Drifting by are an elephant and a mouse, and you give each of them a push of equal strength. The fact that the mouse abruptly changes its path, while the elephant's course is as good as unaltered, is a sure sign that the mass (or, in the language of physics, the inertia or inertial mass) of the elephant is much greater than that of the mouse. Secondly, mass is a measure of how many [atoms](#) there are in a body, and of what type they are. All atoms of one and the same type have the same mass, and adding up all those tiny component masses, the total mass of the body results. Thirdly, in [Newton's theory of gravity](#), mass determines how strongly a body attracts other bodies via the gravitational force, and how strongly these bodies attract it (in this sense, mass is the [charge](#) associated with the gravitational force).

In [special relativity](#), one can also define a mass that is a measure for a body's resistance to changing its motion. However, the value of this [relativistic mass](#) depends on the relative motion of the body and the observer. The relativistic mass is the "m" in Einstein's famous $E=mc^2$ (cf. [equivalence of mass and energy](#)).

The relativistic mass has a minimum for an observer that is at rest relative to the body in question. This value is the so-called *rest mass* of the body, and when [particle physicists](#) talk of mass, this is usually what they mean. Just as in classical physics, the rest mass is a kind of measure for how much matter the body is made up of – with one caveat: For composite bodies, the energies associated with the forces holding the body together contribute to the total mass, as well (another consequence of the [equivalence of mass and energy](#)).

In [general relativity](#), mass still plays a role as a source of gravity; however, it has been joined by physical quantities such as [energy](#), [momentum](#) and [pressure](#).

mass defect

Whenever two or more objects are bound together by strong forces, there is a binding energy – the energy needed to be able to get these objects apart. Since Einstein, we know that [energy](#) and [mass are equivalent](#). To this binding energy there corresponds a mass. It is called the mass defect because, by this amount, the mass of the component object is less than the sum of the masses of its parts.

Some more information about binding energies and the mass defect can be found in the spotlight topic [Is the whole the sum of its parts?](#)

masses in astronomy

While [mass](#) is surely one of the most basic properties of an astronomical object, it is not that easy to actually determine that mass. Most methods utilize the laws of [celestial mechanics](#) to deduce from the way that two (or more) objects orbit each

other their respective masses. In other cases, [relativistic](#) effects such as the [deflection](#) of light or the [Shapiro delay](#) can be used to determine an object's mass.

Synonyms: determination of masses in astronomy

matter

In [general relativity](#): All contents of spacetime that contribute to its curvature: particles, dust, [gases](#), [fluids](#), [electromagnetic](#) and other fields.

In [particle physics](#): All [elementary particles](#) with half-integer [spin](#), such as [electrons](#) and [quarks](#), as well as their composites such as [protons](#) and [neutrons](#), in contrast with [force particles](#).

Max Planck Institute for Astrophysics

Research institute of the [Max Planck Society](#), dedicated to astrophysical subjects such as the evolution of stars, the physics of supernovae, the evolution of galaxies and cosmology. Founded in 1958 in Germany, located in Garching, near Munich.

[MPA website](#)

Synonyms: MPA

Max Planck Institute for Extraterrestrial Physics

Research institute of the [Max Planck Society](#), dedicated to astronomy and astrophysics with observations in the infrared, X-ray and gamma ray part of the electromagnetic spectrum. Founded in 1963 in Germany, located in Garching, near Munich.

[MPE website](#)

Synonyms: MPE

Max Planck Institute for Gravitational Physics/Albert Einstein Institute

See [Albert Einstein Institute](#).

Synonyms: AEI

Max Planck Institute for Radio Astronomy

Research institute of the [Max Planck Society](#), dedicated to infrared and radio astronomy. Founded in 1966, the institute is located in Bonn and Bad Münstereifel-Effelsberg, Germany.

[MPIfR website](#)

Max Planck Society

German organisation dedicated to basic research; operates 84 Max Planck Institutes (as of January 2018) dedicated to research in specific fields of science – see the entries directly above. Founded in 1948 as the successor of the Kaiser Wilhelm Society; administrative headquarters are located in Munich, Germany.

[MPG website](#)

Synonyms: MPG

Maxwell's equations

The four fundamental equations of [electromagnetism](#) that describe how magnetic and electric influences (in physics language: electric and magnetic [fields](#)) are produced: Electric fields are produced whenever there are electric charges or, alternatively, when magnetic fields change over time. Magnetic fields are produced whenever there are electric currents (moving electric charges), but also whenever electric fields change over time. The fact that electric and magnetic fields can exist without the presence of charges or currents, simply by mutual excitation where a change in the magnetic field produces an electric field and vice versa, is the basic phenomenon behind [electromagnetic waves](#).

mechanics

Branch of physics dealing with the motions of objects and how they react to forces acting on them. Depending on the framework used, there is [classical mechanics](#), [relativistic mechanics](#) and [quantum mechanics](#).

mechanics, classical

Synonym: Newtonian mechanics. According to classical mechanics, the movements of bodies are regulated by Newton's three laws of mechanics. The first law states that bodies on which no external force acts stay at rest or move with constant speed along a straight path ("law of inertia"). The second law relates the force acting on a body, the body's [mass](#) and the acceleration caused by the force: Force is equal to mass times acceleration. The third law is the law of "action equals reaction": If a body A acts on a body B with a certain force, then A itself experiences B acting on it with a force that is equal in strength but has the opposite direction.

An alternative version of the second law uses the concept of [momentum](#): The force acting on a body is equal to the change of that body's momentum over time.

Synonyms: classical mechanics

mechanics, relativistic

The generalization of [classical mechanics](#) that takes into account the effects of [special relativity](#). The basic laws are almost unchanged: First of all, bodies on which no external forces act stay at rest or move with constant speed along straight paths – in the language of special relativity: such bodies move on [straight lines in spacetime](#). Secondly: The total force acting on a body is equal to the change of its [momentum](#) over time (but notice: this momentum is defined using the body's

[relativistic mass](#), which depends on the bodies speed relative to the observer).
Thirdly, mass and momentum are conserved quantities – their total sum is the same whenever particles interact (this is equivalent to a slightly modified version of the “action equals reaction” principle of classical mechanics).

There exists an elegant reformulation of these laws of mechanics using four-dimensional concepts adapted to the geometry of spacetime, such as the “four-momentum”.

Synonyms: relativistic mechanics

Mercury

The [planet](#) closest to the sun. In the context of [general relativity](#) it is of interest because, for this planet, the deviation from the orbits of [Newtonian gravity](#) the theory predicts, the relativistic [perihelion shift](#), is especially great. The concord between prediction and observation for this shift constitutes the first successful test of general relativity.

merger

A merger occurs as a consequence of an [inspiral](#), when two [black holes](#) collide to form one single bigger black hole. The emission of [gravitational waves](#) peaks during a merger.

metre

In the international system of units ([SI](#)), the basic unit for length. Since 1983, the official definition uses the constancy of the [speed of light](#) as postulated in [special relativity](#): the metre is defined with the help of the basic unit for time, the [second](#): a metre is the distance that light travels in a vacuum in one 299792458th of a second.

Synonyms: metre

MeV

See [electron volt](#)

Synonyms: Mega-electronvolt

micro

“Micro” as a prefix denotes “one millionths”, making a micro[metre](#) a millionth of a metre.

microquasar

An astronomical object comparable in size to a [star](#), which emits enormous amounts of [energy](#); the processes responsible for the emission are similar to those which happen in [quasars](#) or other [active galactic nuclei](#). The key component of a microquasar is a central [stellar black hole](#); more information about how black holes can lead to such enormous energy output can be found in the spotlight text [Luminous disks: How black holes light up their surroundings](#).

microwaves

[Electromagnetic radiation](#) with [wave-lengths](#) between a millimetre and 30 centimetres, corresponding to [frequencies](#) between some and some hundred billion oscillations per second.

Looking up to the night sky, almost all energy that reaches us of the [cosmic background radiation](#) is in the form of microwaves.

Milky Way

1. Our home [galaxy](#) – a spiral galaxy, a disk of stars with a diameter of roughly hundred thousand [light-years](#) and a thickness between three- and six thousand [light-years](#), containing about 100 billions of stars. It also contains a [supermassive black hole](#) in its centre – more about that in our spotlight topic [The black heart of the Milky Way](#).

2. As our sun is located within the disk of the Milky Way galaxy, there are directions in which we can observe comparatively few of the other stars in our galaxy (namely as we look perpendicularly to the disk, or nearly so) and directions in which we can see a large amount of stars (as we look approximately in parallel with the disk plane). The result is that there is a dim band in the night sky marking the disk plane (the directions where we see a lot of the other stars). This band is also known as the Milky Way.

milli

“Milli” as a prefix denotes “one thousandth”, making a [millimetre](#) a thousandth of a metre.

minute of arc

See [arcminute](#), [arcsecond](#).

molecule

Composite object consisting of two or more [atoms](#), bound together by [electromagnetic](#) forces.

Synonyms: molecules

momentum

Physical quantity that is equal to a body’s [mass](#) times its [velocity](#) (in [special relativity](#): its [relativistic mass](#) times its [velocity](#)).

What makes momentum a useful quantity is that it is conserved – if several bodies interact, the sum of their momenta before and after the interaction is the same. Momentum is neither created nor destroyed, merely passed on from one body to another.

In [general relativity](#), momentum is one of a number of quantities (like [mass](#) and [energy](#)) that acts as a source of gravity.

muon

An [elementary particle](#) which is a somewhat heavier version of the [electron](#). Its [electric charge](#) is the same as that of the electron, and like that particle, it does not interact via the [strong force](#). The muon's [mass](#) is about 207 times that of the electron.

naked singularity

In [general relativity](#), [singularities](#) – environs of infinite curvature – that form in the collapse of massive bodies or in similar processes are typically hidden inside [black holes](#), in other words: spacetime in their vicinity is distorted so much that no information about the singularity can ever reach the outside world. Hypothetical singularities which are not cloaked in this way, and thus are visible to the rest of the cosmos, are called “naked”. By the [cosmic censorship](#) hypothesis, no realistic kind of collapse can lead to the formation of a naked singularity.

nano

“Nano” as a prefix denotes “one billionth”, making a nanometre one billionth of a metre.

National Aeronautics and Space Administration (NASA)

Part of the US government in charge not only of manned space missions, but also responsible for numerous highly successful satellite and probe missions. NASA is a partner in projects such as the [Hubble space telescope](#) or the gravitational wave detector [LISA](#).

[NASA website](#)

Synonyms: NASA

National Radio Astronomy Observatory

US national institute for [radio astronomy](#), located in Charlottesville, Virginia. Responsible for operating the Very Large Array of radio telescopes in New Mexico and the Very Large Baseline Array, an array of ten far-apart [radio telescopes](#).

[NRAO website](#)

Synonyms: NRAO

neutrino

Type of [elementary particle](#) that is related to the [electron](#), but carries no electric charge and has an extremely small mass. There are three types of neutrinos, called electron-neutrino, muon-neutrino and tau-neutrino.

Synonyms: neutrinos

neutron

Particle that is electrically neutral and comparatively massive; the atomic [nuclei](#) consist of neutrons and [protons](#).

Neutrons are not [elementary particles](#), they are compound particles consisting of [quarks](#) that are bound together through the [strong nuclear interaction](#). Collectively, neutrons, protons and a number of similar particles are called [baryons](#).

[Neutron stars](#) are mainly made of neutrons.

Synonyms: neutrons

neutron star

Final stage of massive [stars](#) that explode as a [supernova](#). In the explosion process, the core of the star collapses to form a compact object with roughly 1.4 [solar masses](#) that mostly consists of [nuclear](#) matter, predominantly of [neutrons](#).

For astronomers, neutron stars are of interest as there exists a variety called [pulsars](#) from which they receive highly regular pulses of [electromagnetic radiation](#). For relativists, they are interesting as the typical effects of [general relativity](#) are very pronounced in objects that compact (compare [PSR 1913+16](#), double pulsar [PSR J0737-3029A/B](#)).

Newton

English physicist of the 16th/17th century, see [Newtonian gravity](#), below.

In the international system of units ([SI](#)) the unit of force, abbreviation N. One Newton is the force needed to impart on a body of mass one [kilogram](#) an [acceleration](#) of one metre per second-squared.

Newton's constant

See [gravitational constant](#)

Newtonian gravity

In pre-Einstein mechanics, which goes back to the English physicist and mathematician Isaac Newton (1643-1727), gravity is a force with which masses act on each other. As other forces do, they cause bodies to accelerate.

In its simplest form, Newton's law of gravity describes the force acting between two spherical, symmetric masses: The force with which the first sphere acts on the second is equal to the [mass](#) of the first sphere times the mass of the second sphere

times Newton's [gravitational constant](#), divided by the square of the distance between the centre-points of the two spheres.

How to remove from this law more complicated gravitational effects, see the article [The gravitation of gravitation](#). The differences between Newton's gravitation and Einstein's theory of gravitation, the theory of General Relativity, can be described systematically in the frame of the so called post-Newtonian approximation.

Synonyms: Newton's law of gravity

nitrogen

Chemical element whose [atoms](#) have seven [protons](#) each in their [nuclei](#).

In the context of [relativity](#), more concretely: [cosmology](#), nitrogen is interesting as an indicator of [chemical evolution](#): Its nuclei are not produced during [Big Bang Nucleosynthesis](#), but they are produced by [nuclear fusion](#) reactions in the interior of [stars](#). The presence of nitrogen (or, for that matter, of other elements such as [oxygen](#) or iron) in an astronomical object is an indicator that stellar fusion has taken place, and hence that the abundances of the different elements do not reflect the element abundances in the early universe.

no-hair theorem

See [black hole uniqueness theorems](#).

nonlinear

In some physical theories, influences can simply be added up – take the [electric force](#) associated with one particular [charged](#) body, the electric force associated with a different body, and their sum will be the electric force felt by a [test particle](#) when both bodies are present. Such theories are called linear; theories where separate contributions do not simply add up are called nonlinear, an important example being Einstein's [general relativity](#).

Synonyms: nonlinearity

nuclear fission

Processes in which a heavier atomic [nucleus](#) splits up into several lighter nuclei. If the initial nucleus is heavy enough, energy is set free in the split. Nuclear fission reactions are used in nuclear reactors to produce electrical energy, and in nuclear weapons to power an energetic explosion.

Some more information about nuclear fission can be found in the spotlight topic [Is the whole the sum of its parts?](#). The role played in nuclear fission and fusion by Einstein's famous formula $E=mc^2$ is the subject of the spotlight topic [From \$E=mc^2\$ to the atomic bomb](#).

nuclear force

Force that binds [protons](#) and [neutrons](#) together to form atomic [nuclei](#); a side-effect of the [strong interaction](#).

See also [strong interaction](#) and [weak interaction](#).

nuclear fusion

Processes in which two lighter atomic [nuclei](#) merge to form a more massive nucleus. For nuclei lighter than those of iron, energy is released in fusion. This is the main source of energy of ordinary [stars](#) like our [sun](#).

Some more information about nuclear fusion can be found in the spotlight topic [Is the whole the sum of its parts?](#). The role played in nuclear fusion and fission by Einstein's famous formula $E=mc^2$ is the subject of the spotlight topic [From \$E=mc^2\$ to the atomic bomb](#).

nuclear physics

That branch of physics dealing with the properties of atomic [nucleus](#). One connection to relativistic physics is the fact that nuclear physics is needed to describe the properties of matter in the early universe of the [big bang models](#) and in the interior of [neutron stars](#).

nucleosynthesis

See [Big Bang Nucleosynthesis](#).

nucleus

Extremely dense central region of an [atom](#), consisting of [protons](#) and [neutrons](#) held together by [nuclear forces](#). The number of protons determines what [chemical element](#) a nucleus represents.

Typical diameter for atomic nuclei are in the region of a quadrillionth of a metre=[10⁻¹⁵](#) metres. This makes nuclei about a hundredth of a thousandth as large as atoms.

Synonyms: nuclei

numerical relativity

Subdiscipline of physics devoted to the use of computer simulations for exploring the structure and consequences of Einstein's theories, [special](#) and [general relativity](#).

Notably, the centerpiece of general relativity are [Einstein's equations](#), which relate certain properties of the [matter](#) contained in a [spacetime](#) to that spacetime's [geometry](#). A model universe in which matter distorts the geometry – and is in turn influenced by those distortions – in exactly the way prescribed by Einstein's equations is called a solution of these equations. Some simple solutions can simply be written down on a piece of paper (“exact solutions”). More complicated situations can only be described by simulating space, time and matter in a computer (“numerical solution”), and this is one of the main tasks of numerical relativity.

Numerical relativity has led to interesting results about [black holes](#) and [gravitational waves](#), for instance about the gravitational wave produced when two black holes collide and merge. They have also shed light on what general relativity predicts for the properties of spacetime close to a black hole's central [singularity](#) (further

information about this can be found in the spotlight text [Of singularities and breadmaking](#)). The branch of numerical relativity that is of interest for the study of phenomena such as [supernovae](#), [jets](#), and merging or collapsing [neutron stars](#) is relativistic (magneto-)hydrodynamics, more about which can be found in the spotlight text [The realm of relativistic hydrodynamics](#).

observer

In the context of relativity, “observer” can mean two different things.

Often, observer is synonymous with [reference frame](#) or (spacetime-) [coordinate system](#): An observer in this sense is someone who assigns coordinates to everything that happens around him. In particular, all events are assigned space coordinate values and a time coordinate value. In the context of [special relativity](#), it is often the case that when one talks about an observer, what is really meant is an [inertial observer](#), corresponding to a special type of reference frame.

On other occasions, the term is used in a more narrow sense – in those cases, an observer is someone sitting at a certain point in space and using the [light](#) signals reaching that location to construct an image of his surroundings. In the context of optical effects in relativity, for instance [gravitational lensing](#), observer is usually meant in this way.

Olbers' paradox

In an infinitely extended universe that does not change over time and is evenly filled with stars, the “night sky” would look as bright as the surface of the sun. The reason: The farther away a star, the weaker the light we receive from it. But: The greater the distance, the greater the number of stars that have exactly that distance from us. In an eternal and infinite universe, the two effects cancel exactly.

The [big bang models](#) based on Einstein's [general theory of relativity](#), with a changing universe evolving out of a hot initial state, make no such counter-factual prediction.

oxygen

Chemical element whose [atoms](#) have eight [protons](#) each in their [nuclei](#).

In the context of [relativity](#), more concretely: [cosmology](#), oxygen is interesting as an indicator of [chemical evolution](#): Oxygen nuclei are not produced during [Big Bang Nucleosynthesis](#), but they are produced by [nuclear fusion](#) reactions in the interior of [stars](#). The presence of oxygen in an astronomical object is an indicator that stellar fusion has taken place, and that the abundances of the different elements thus do not reflect the element abundances in the early universe.

parsec

Astronomical unit of distance:

1 parsec = 3.26 [light-years](#)
= 200 000 [astronomical units](#)
= 20 trillion miles
= 30 trillion kilometres.

Abbreviation: pc. Parsec is an acronym for “parallax second”.

particle accelerator

The most important experimental technique of [particle physics](#): accelerating electrically [charged](#) particles with the help of [electric forces](#), make them collide with each other and, from the result of the collision, draw conclusions about the properties of elementary particles and their interactions.

It is an intriguing possibility, suggested by models based on the ideas of [string theory](#), that particle accelerators such as the [LHC](#) might actually produce miniature [black holes](#) (for more about this, see the spotlight text [Particle accelerators as black hole factories?](#)).

particle physics

The branch of physics that deals with particles that are, to the best of today’s knowledge, not made up of more fundamental sub-units, for instance with [electrons](#), [quarks](#) or [neutrinos](#). Also included is the study of some species of particles that do have more elementary constituents, such as [protons](#) or [neutrons](#), but not of larger systems such as atomic [nuclei](#) (that would be [nuclear physics](#)) or, even worse, whole [atoms](#). Questions such as, whether the particles nowadays thought to be elementary really are elementary, or are, for instance, different manifestations of one and the same species of [string](#), fall within the scope of particle physics.

The theoretical tools of particle physics are the so-called [quantum field theories](#) which allow the description of elementary particles on the basis of both [quantum theory](#) and [special relativity](#), while the main experimental tools are [particle accelerators](#) in which particles are accelerated and then brought to collision.

path integrals

Synonym: Sum over histories. A technique for performing calculations in [quantum theory](#). Roughly speaking, the probability for a certain outcome (for instance, a particle reaching location A at time t) is calculated by performing a sum over all possible ways in which this particular outcome can come about.

A description of the path integral formulation of quantum theory can be found in the spotlight text [The sum over all possibilities](#).

Pauli exclusion principle

Basic principle of [quantum theory](#) stating that no two [fermions](#) can be in exactly the same state – for instance: no two fermions with identical properties can be at the same location. Formulated by the physicist Wolfgang Pauli.

[Electrons](#) are fermions, and the Pauli exclusion principle plays a crucial role in bringing about the properties of matter as we know them: It is responsible for the fact that the electrons of [atoms](#) do not all cluster together in the lowest-energy state close to the atomic [nucleus](#), but instead spread out, occupying different states. This is what gives atoms their shell structure, responsible for different atoms' different chemical properties.

perihelion

For a [planet](#) or other heavenly body orbiting the [sun](#) on an [elliptic orbit](#), that point of the orbit closest to the sun. In the context of [general relativity](#), the perihelion is of great interest as that theory predicts a slight motion of this point around the sun, cf. [\(relativistic\) perihelion shift](#).

perihelion advance, relativistic

For planetary orbits, there is a small difference between the predictions of [Newtonian gravity](#) and [general relativity](#). For instance, in Newton's theory, the orbital curve of a lonely planet orbiting a star is an [ellipse](#). In general relativity, it is a kind of rose or rhodonea curve. Such a curve is similar to an ellipse curve, which shifts a bit with each additional orbit. The shift can be defined by looking at the point which is closest to the sun ([perihelion](#)) on each orbit. The additional relativistic shift is, hence, called relativistic perihelion shift or relativistic perihelion advance. A picture can be seen on the page [A planet goes astray](#) in the chapter [General relativity](#) of [Elementary Einstein](#).

Synonyms: perihelion shift, relativistic

Perimeter Institute for Theoretical Physics

Privately funded institute for basic research in theoretical physics, located in Waterloo, Canada. Currently, the main areas of research are quantum computing, the foundations of [quantum theory](#), and [quantum gravity](#).

[Perimeter Institute website](#)

photo effect

When light shines onto a metal, it can knock [electrons](#) out of the metal's [atoms](#). This is the photoelectric effect, and its properties – how does the number and [energy](#) of the electrons depend on the [frequency](#) and intensity of the light? – can only be explained if one accepts that light is no mere [electromagnetic wave](#), but somehow made up of some kind of light particles. With this postulate, Einstein, in 1905, paved the way for the later development of [quantum mechanics](#).

Synonyms: photoelectric effect

photon

Synonym: light particle, light quantum. In [quantum theory](#), light is not a continuous [electromagnetic wave](#), but a steady stream of tiny energy packets – photons.

Synonyms: photons

photon radius

In a certain distance from a spherically symmetric [black hole](#), the [deflection of light](#) because of the black hole's gravity is so great that light can move on closed circular orbits – [photons](#) (light particles) can, at this distance, orbit the black hole like a [planet](#) the sun. This particular distance is called the photon radius.

An observer at rest at this distance can see the back of his or her own head (or at least a small region thereof), as the photons emitted by the back of the head travel once around the black hole and fly directly into his or her eyes.

Pierre Auger Observatory

An observatory in western Argentina built to study high energy [cosmic rays](#). From the viewpoint of relativity, one interesting aspect of this is the possibility that such cosmic rays might produce miniature [black holes](#) (see the spotlight text [Particle accelerators as black hole factories?](#)).

[Auger Observatory website](#)

Planck energy

Natural unit of energy that can be obtained by combining the fundamental natural constants that govern spacetime, the strength of gravity and the quantum world: the [gravitational constant](#), [Planck's constant](#) and the [speed of light](#). Whenever [elementary particles](#) reach this kind of energy, in addition to the effects of [quantum theory](#), the effects of [general relativity](#) should become important, in short: such situations could only be described adequately using a theory of [quantum gravity](#).

Compare: [Planck length](#), [Planck time](#), [Planck mass](#).

Planck length

Natural length that can be obtained by combining the fundamental natural constants that govern spacetime, the strength of gravity and the quantum world: the [gravitational constant](#), [Planck's constant](#) and the [speed of light](#). It amounts to roughly $1.6 \cdot 10^{-35}$ meters [see [exponential notation](#)].

At such length scales, both the effects of [quantum theory](#) and those of [general relativity](#) should become important, in short: whatever concerns such scales can only be described adequately using a theory of [quantum gravity](#).

Compare [Planck time](#), [Planck energy](#), [Planck mass](#).

Planck mass

Natural unit of mass that can be obtained by combining the fundamental natural constants that govern spacetime, the strength of gravity and the quantum world: the [gravitational constant](#), [Planck's constant](#) and the [speed of light](#). Compared with the masses we're used to in everyday life, the Planck mass is rather small, a mere 2 hundredth of a thousandth of a gram. However, if this mass is concentrated in a single [elementary particle](#) then, in addition to the effects of [quantum theory](#), the effects of [general relativity](#) should become important, in short: such a particle could only be described adequately using a theory of [quantum gravity](#).

Compare [Planck length](#), [Planck time](#), [Planck energy](#).

Planck time

Natural interval of time that can be obtained by combining the fundamental natural constants that govern spacetime, the strength of gravity and the quantum world: the [gravitational constant](#), [Planck's constant](#) and the [speed of light](#). It amounts to about $5 \cdot 10^{-44}$ [seconds](#) [see [exponential notation](#)] and is the time it takes light to traverse one [Planck length's](#) worth of distance.

At such time scales – for instance: at [cosmic time](#) comparable to the Planck time in the [big bang models](#) – both the effects of [quantum theory](#) and those of [general relativity](#) should become important, in short: such time intervals and what happens in them can only be described adequately using a theory of [quantum gravity](#).

Compare: [Planck length](#), [Planck energy](#), [Planck mass](#).

Planck units

Natural units for length, time, energy and mass, obtained by combining the fundamental natural constants that govern spacetime, the strength of gravity and the quantum world: the [gravitational constant](#), [Planck's constant](#) and the [speed of light](#).

See: [Planck length](#), [Planck time](#), [Planck energy](#), [Planck mass](#).

Planck's constant

Fundamental constant of [quantum theory](#); of the dimension energy times time. For instance, the energy of a single [photon](#) is equal to Planck's constant times the photon's [frequency](#). Abbreviated as h in formulae.

Planck's radiation law

The fundamental law governing the properties of the simplest form of [thermal radiation](#) – that emitted by a [blackbody](#). It describes the [spectrum](#) of such radiation in terms of universal constants and a single parameter – the body's [temperature](#). The result is also called a blackbody spectrum.

plane

A [surface](#) within which the axioms of [Euclidean geometry](#) (synonym: plane geometry) hold – the rules of geometry as they are taught in high school, with well-known formulae such as [Pythagoras's theorem](#) and “the perimeter of a circle is 2 times pi times its radius” hold.

planet

Planets are not-too-small companions of a [star](#) that are not stars themselves (nor ever were stars). In our [solar system](#), the planets are, listed from the one closest to the sun to the one farthest: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. As of August 2006, Pluto, which used to be a proper planet, is officially a “dwarf planet”. In the night sky, the distinguishing characteristic of planets is that they move around relative to the unchanging background of stars – which gave them their name, loosely translated from Greek as “wanderers”.

plasma

[State](#) of matter in which the [atoms](#) are largely or even completely separated into [electrons](#) and atomic nuclei, which fly around in a highly energetic mixture.

Compare the other [states of matter](#): [solid state](#), [gas](#), [liquid](#).

point

Elementary “building block” of geometrical entities such as [surfaces](#) or more general [spaces](#). For instance, a surface is the set of all its points, of all possible locations on the surface, and all geometrical objects in that surface are defined by the points that belong to them – for instance, a line on the surface is the set of (infinitely many) points.

polarization

[Waves](#) that are especially simple can be completely described by stating the direction in which they propagate, their [speed](#) of propagation, [frequency](#), and [amplitude](#). But there are also simple waves where these quantities are not sufficient for a complete description – for these waves, the oscillation has an orientation in space. This orientation, which is called polarization, needs to be specified as well.

For example, for [electromagnetic waves](#), the polarization describes the directions of the [electric](#) and the [magnetic](#) fields. For [gravitational waves](#), the polarization describes the orientation of the two orthogonal directions in which distances are maximally stretched and squeezed as the gravitational wave passes.

positron

Positrons are the [anti-particles](#) of [electrons](#): light [elementary particles](#) with positive electric [charge](#).

Synonyms: positrons

post-Newtonian formalism

For situations in which [gravity](#) is very weak, [general relativity](#) and [Newton’s theory of gravity](#) lead to very similar predictions for the motion of bodies (e.g. the planets in our solar system) and the propagation of light. Such situations can be described by starting out with the Newtonian description and then, step by step, adding correction terms that take into account the effects of general relativity. The post-Newtonian formalism is a method for performing those step-by-step corrections. As the correction terms are ordered in a systematic way (the largest effects are called “of first post-Newtonian order, 1pN”, the next smallest ones of second order, and so on), the progression of ever smaller corrections is also called the post-Newtonian expansion.

The post-Newtonian formalism is also crucial to describe relativistic effects in binary systems (binary [neutron stars](#) or [black holes](#)). It therefore plays a key role in the direct detection of gravitational waves: Based on the post-Newtonian formalism, various possible gravitational waveforms are modeled and used for searching the [detector](#) data for signals. The post-Newtonian formalism is a vital ingredient for all waveform models, but is particularly important for binary neutron stars. For those

systems, most of the observed signal is in the regime where post-Newtonian terms are dominant. It was thanks to a post-Newtonian model that [gravitational waves from the merger of two orbiting neutron stars](#) were detected for the first time in 2017.

Synonyms: post-Newtonian expansions

potential energy

When an object is being acted upon by a force like the [electric](#) or [gravitational](#) force, then it can be assigned an [energy](#) that depends only on its location relative to the source of the force. This energy is called the potential energy – “potential” as it can easily be transformed into [kinetic energy](#), energy associated with the object’s motion: As the object yields to the pull or push of the force, its potential energy decreases while the energy associated with its motion increases.

pressure

A measure for the strength of the resistance with which matter (for instance a [gas](#)) resists attempts to decrease the volume it occupies.

primordial

In [cosmology](#): At, from, or relating to the beginning (or at least the early phases) of the universe. For example, the primordial abundances of the chemical [elements](#) are the abundances right after [Big Bang Nucleosynthesis](#), at a [cosmic time](#) of a few minutes.

proton

Particle that carries positive electric [charge](#) and is comparatively massive; the atomic [nuclei](#) consist of protons and [neutrons](#).

Protons are not [elementary particles](#), they are compound particles consisting of [quarks](#) that are bound together through the [strong nuclear interaction](#). Collectively, protons, neutrons and a number of similar particles are called [baryons](#).

Synonyms: protons

PSR1913+16

A specific [binary system](#) consisting of two orbiting [neutron stars](#), one of which is a [pulsar](#) from which we here on Earth receive regularly spaced radio pulses. From the point of view of [general relativity](#), the system is interesting not only because, using the pulses, one can measure effects such as [Shapiro delay](#) with impressive precision, but because it has given us the first indirect proof for the existence of [gravitational waves](#): the orbital period of the two stars becomes slightly shorter with each orbit, exactly in the way predicted by [general relativity](#) due to the energy the system radiates away in the form of gravitational waves.

pulsar

Rotating [neutron star](#) from which regular pulses of radiation reach the Earth. Behind those pulses is the fact that the pulsar sends out narrowly focussed beams of radiation that, due to the pulsar's rotation, sweep through space like the beam of a light-house. An animation illustrating this effect can be found on the page [Neutron stars and pulsars](#) in the chapter [Black holes & Co.](#) of [Elementary Einstein](#).

Synonyms: pulsars

Pythagoras's theorem

For a [triangle](#) in the [plane](#) (or in a more general [flat space](#)), two of whose sides form an angle of 90 degrees, the following holds: The lengths a and b of the two sides that form the 90 degrees angle and the length c of the third side are related by the formula

$$a^2+b^2=c^2.$$

quantize

First of all, quantization is the process describing the transition from a [classical](#) theory to the corresponding [quantum theory](#). For instance, if you quantize classical [electrodynamics](#), you will end up with its quantum version, [quantum electrodynamics](#).

On the other hand, to be quantized means for a physical quantity to be divided into little building blocks or packets. For instance, in [quantum theory](#), the [energy](#) of [light](#) is quantized: a given quantity of light consists of a finite number of energy packets called [photons](#).

Synonyms: quantization

Quantum chromodynamics

[Quantum theory](#) of the [strong nuclear](#) interactions between [quarks](#) (or compound particles made of quarks). These interactions are described via the exchange of [carrier particles](#) called [gluons](#), for instance: when two quarks attract each other via the strong nuclear force, that influence is transmitted by gluons flying back and forth between them.

Synonyms: QCD

quantum cosmology

According to the [big bang models](#), the [energy density](#) of the very early universe was extremely high, with the contents of the observable universe compressed into a volume much smaller than that of an atomic [nucleus](#). Under such circumstances, the effects of [quantum physics](#) and of [general relativity](#) should become equally important, in other words: This era of our cosmic evolution should be described using a theory of [quantum gravity](#). Quantum cosmology encompasses all attempts to apply various candidate theories of quantum gravity to the physics of the early universe, and to describe the universe as a whole as a quantum system.

More information can be found in the spotlight topic [Searching for the quantum beginning of the universe](#). Some quantum cosmological applications of [loop quantum gravity](#) can be found in the spotlight texts [Avoiding the big bang](#) and [Taming infinities with loops](#).

quantum effects

All effects and phenomena that follow from the fact that, deep down, our world obeys not the laws of [classical](#) physics, but those of [quantum theory](#). Examples are [tunneling](#) and consequences of the [Pauli exclusion principle](#) for the structure of [atoms](#).

Quantum electrodynamics

[Quantum theory](#) of [electromagnetic](#) interactions. These interactions are described via the exchange of [carrier particles](#) called [photons](#), for instance: when two electrons repel each other via the electromagnetic force, that influence is transmitted by photons flying back and forth between them.

Quantum electrodynamics is one facet of the [standard model of particle physics](#). It is also the simplest example for a [relativistic quantum field theory](#) – a theory that is both a [quantum theory](#) and based on the principles of [special relativity](#).

Synonyms: QED

quantum field theory, relativistic

Collective name for theories that are [quantum theories](#) based on the principles of [special relativity](#). Typically, in relativistic quantum theory there exists for every species of particle a corresponding species of [anti-particle](#); forces are transmitted by the exchange of [carrier particles](#).

The simplest example of a relativistic quantum field theory is [quantum electrodynamics](#).

quantum gravity

Theory based both on the effects, concepts and laws of [quantum theory](#) and on those of [general relativity](#). To date, no complete such theory exists; the best-known candidate theories are [string theory](#) and [loop quantum gravity](#).

Some information on the question of quantum gravity can be found in the chapter [Relativity and the quantum](#) of [Elementary Einstein](#), starting with the page [Border regions of gravity](#).

Selected aspects of quantum gravity are described in the category [Relativity and the quantum](#) of our [Spotlights on relativity](#).

quantum mechanics

In a more general sense: synonymous with [quantum theory](#). In a more restricted sense: The quantum theory of particles moving under the influence of forces – wherein the particles are described as quantum objects, while the forces are not. An important application of quantum mechanics is the physics of the electron shells of

[atoms](#) (“atomic physics”, for short). Attempts to extend the quantum laws to govern the forces themselves lead to [relativistic quantum field theories](#).

quantum particles

In [classical](#) physics, one can picture particles as little clumps of matter. At every time, such a clump has a definite location in space. In [quantum theory](#), on the other hand, (quantum) particles are much more elusive. Their most complete description involves an abstract “state” that allows one to calculate probabilities, in particular: how likely it is to detect the particle, at a given time, at a given location.

quantum physics

Quantum physics comprises all theories, models, experiments and applications that are based on the laws of [quantum theory](#).

quantum theory

The framework for formulating the physical laws that govern the world at microscopic length-scales – the physics of the micro-world, for instance of [atoms](#), atomic [nuclei](#) or [elementary particles](#), but also the physics of ultra-precise measurements such as those made by [gravitational wave detectors](#).

The laws of quantum theory are fundamentally different from our everyday experience and from those of [classical](#) physics.

The first unusual feature is that, in many cases, quantum theory merely allows statements about *probabilities*. For instance, in classical physics, one can assign to every particle, at every point in time, a location and a velocity. Whosoever can measure those quantities precisely can, in principle, predict where the particle in question can be found at every point in the future. In quantum theory, all one can assign to a system of particles is an abstract quantum state from which can be derived no precise predictions, but merely the probabilities for detecting a specific particle at a given time in a given place. Whether or not one will really find the particle at that location is governed by chance.

The second unusual feature is a fundamental restriction placed on the exactness of certain measurements ([Heisenberg uncertainty relation](#)). For instance, the more precise one measures a particle’s location, the less definite any statements one can make about its velocity.

The third feature is how quantum theory came by its name: A number of physical quantities in nature come in little packets, in quanta. For instance, according to quantum theory, [electromagnetic radiation](#) is made of tiny quanta of energy called [photons](#).

Examples for quantum theories are [quantum mechanics](#) and [relativistic quantum field theories](#) such as [Quantum electrodynamics](#) or other parts of the [standard model of particle physics](#).

quark

Elementary particle that is subject to the [strong interaction](#). Quarks come in six species: up, down, strange, charme, bottom and top (with the last two sometimes called beauty and truth).

Quarks are the constituents of [protons](#) and [neutrons](#), and thus the material atomic [nuclei](#) are made of.

Synonyms: quarks

quark-gluon-plasma

An exotic form of matter which was in all probability present in the early universe shortly after the [big bang](#). Under ordinary circumstances, [quarks](#) only occur inside larger particles, such as [protons](#) or [neutrons](#), tightly bound together as they are by the [carrier particles](#) of the [strong nuclear force](#), which are called [gluons](#). However, at extremely high [densities](#) and [temperatures](#) it is thought that these larger particles break up, creating a dense and strongly interacting soup of quarks and gluons – a quark-gluon-plasma.

It should be possible to create a quark-gluon-plasma artificially at [particle accelerators](#); in fact, there are strong indications that particle physicists have managed to do just that using the [Relativistic Heavy Ion Collider](#).

quasar

Class of [active galactic nuclei](#). First noticed by [radio astronomers](#) as exceedingly bright radio sources that, in the night sky, did not appear more extended than ordinary stars – thus their name, a contraction of quasi-stellar radio source.

Synonyms: quasars

quasi-normal modes

Quasi-normal modes are a mathematical description of the [energy](#) dissipation of a perturbed [black hole](#) while it settles towards stationary configuration, which corresponds to its [ringdown frequencies](#).

radar

Abbreviation for “radio detection and ranging” – how to detect objects, and measure their distance, by means of sending out radio waves and detecting their reflection. Widely used in air and sea traffic; for [general relativity](#), radar is of interest as radar signals reflected by planets can be used to measure the [Shapiro delay](#) – the fact that it takes those signals passing close to the sun a little longer to reach us than expected by classical physics.

radiation

In a general sense: Collective name for all phenomena in which energy is transported through space in the form of waves or particles. In a more restricted sense, the word is often used synonymously with [electromagnetic radiation](#).

radio astronomy

Branch of astronomy dedicated to the detection and evaluation of [radio waves](#) from outer space. Such observations have led to the discoveries of [radio galaxies](#), [quasars](#) and [pulsars](#).

radio galaxy

One variety of the [active galactic nuclei](#) of young [galaxies](#), whose central region radiates extremely great amounts of energy. Radio galaxies are distinguished by radiating off extremely high energies in the form of radio waves (more than 10^{35} Watts), from sources that are often located outside the visible part of those galaxies. Usually, the sources are radio bubbles, huge regions of gas whose radiation is stimulated by [jets](#).

To the best of current knowledge, what's behind all that energy output is a [supermassive black hole](#) in the galactic core.

radio signals

See [radio waves](#)

radio telescope

Any antenna used for [radio astronomy](#), for instance to observe [pulsars](#) or [radio galaxies](#).

radio waves

Variety of [electromagnetic radiation](#) with [frequencies](#) of a few thousand to a few billion oscillations per second, corresponding to [wave-lengths](#) of a few kilometres to a few centimetres. True to their name, these are the electromagnetic waves that bring radio and TV programs from the broadcast towers to our personal antennas and receivers. Cosmic radiowaves also make for interesting observations – see [radio astronomy](#).

Synonyms: radio signals

recombination

In the early universe as described by the [big bang models](#): transition that occurs at [cosmic time](#) around 380 000 years. At this point, the universe has cooled down sufficiently for atomic [nuclei](#) and [electrons](#) to form [atoms](#) – without immediately being ripped apart by electromagnetic radiation. The etymology of recombination does not quite correspond to the physics – this is not a re-combination, it is the first such combination in the universe as we know it. In the process, the [cosmic background radiation](#) decouples from the material universe.

Synonyms: recombination phase

redshift

The frequency of a simple [light wave](#) is directly related to its colour (cf. [spectrum](#)). For the lowest frequencies of visible light, that colour is red, light of the highest frequencies appears blue. If the frequency of a light wave is shifted towards lower frequencies (for instance by the [doppler shift](#)), that corresponds to a colour shift towards the red end of the spectrum, and is hence called a redshift. Consequently, a shift towards higher frequencies is called blueshift.

From this, “redshift” has come to acquire a more general meaning. It is used to denote *any* shift towards lower frequencies, even for types of electromagnetic radiation where the frequencies do not correspond to any visible colour, and more generally still, for other types of waves as well (for instance for [gravitational waves](#)).

In the context of [general relativity](#), the [gravitational](#) and the [cosmological redshift](#) are of particular interest.

redshift, gravitational

In [general relativity](#), light moving away from a mass or other source of gravity experiences a shift towards lower [frequencies](#). This is called the *gravitational redshift*; on the other hand, light falling towards a mass is blueshifted. Both effects together are called the *gravitational frequency shift*. The gravitational redshift is closely related to gravitational [time dilation](#).

Information about an astrophysical application of this effect can be found in the spotlight text [Gravitational redshift and White Dwarf stars](#).

reference frame

Already in [special relativity](#), motion is relative, and whenever there is talk about a moving clock, one must give the additional information: Moving relative to whom or what? Such a “whom or what”, in other words: An object together with a recipe to determine locations relative to that object and to measure time, is called a reference frame.

In special relativity, there exists a special and very important class of reference frame, so-called inertial reference frames, in short: [inertial frames](#).

relativistic

Models, effects or phenomena in which [special relativity](#) or [general relativity](#) play a crucial role are called relativistic.

Examples are relativistic [quantum field theories](#) as theories based on special relativity, or the relativistic [perihelion shift](#) as a consequence of general relativity.

In addition, conditions under which the difference between relativistic physics and ordinary, [classical](#) physics are especially pronounced, are also called relativistic. For instance, when material objects reach speeds close to [speed of light](#), one talks of relativistic speeds, while speeds that are so small compared to light as to make relativistic effects undetectably small are non-relativistic.

Relativistic Heavy Ion Collider

A [particle accelerator](#) operated by [Brookhaven National Laboratory](#) on Long Island, New York. It brings heavy [ions](#) – the [nuclei](#) of [atoms](#) that have been stripped of all their [electrons](#) – into collision at high [energies](#); the resulting states of matter give valuable information about the very early high temperature universe, and about the physics that should be included in the [big bang](#) models of relativistic cosmology.

[RHIC webpages](#)

Synonyms: RHIC

relativistic mass

One prediction of [special relativity](#) is that, the faster an object already is, the more difficult it is to accelerate it even further. One consequence of this is that it is impossible to accelerate a material object to the speed of light: The faster the object already is, the more force has to be used to increase its speed, and close to the speed of light, this effect becomes so strong that, finally, one would have to use infinite force to effect the final, decisive acceleration.

Traditionally, in classical physics, the resistance of an object to changes of its state of motion is its (inertial) [mass](#). The relativistic mass of an object is defined in the same way, and the value an observer measures for this relativistic mass increases as an object moves faster and faster relative to that observer.

relativity principle

Basic principle of [special relativity](#): for two observers moving relative to each other with constant relative velocity (more specifically: for two [inertial observers](#)) the laws of physics are the same. There is no key experiment by which one could argue that one of the observers is “at rest” in an absolute sense – as far as physics is concerned, all such observers are equal, no one has more right than another to regard himself as being at rest, and motion (at least: motion with constant velocity) is defined only as relative motion of observers with respect to each other.

For an introduction to some of the consequences Einstein derived from the relativity principle, check out the chapter [special relativity](#) of [Elementary Einstein](#).

relativity theory

The modern theories of space and time that go back to Albert Einstein: His [special theory of relativity](#), which ignores the effects of [gravitation](#), and his [general theory of relativity](#), in which gravitation is included as a distortion of space and time.

For an introduction to the basics of both theories of relativity, check out the chapters [Special relativity](#) and [General relativity](#) in [Elementary Einstein](#).

Synonyms: theory of relativity, theories of relativity

resonant detector

Detector for [gravitational waves](#) in which it is attempted to measure the influence of these waves on an oscillating test mass.

More information about how such detectors work can be found in the spotlight topic [Small vibrations](#).

rest mass

In [special relativity](#), the (inertial) [mass](#) of an object depends on how fast that object moves relative to the observer. The rest mass is the inertial mass of an object, measured by an observer relative to whom the object is at rest. With this definition, the rest mass is a kind of measure for how much matter is contained in a body.

RHIC

See [Relativistic Heavy Ion Collider \(RHIC\)](#).

Ricci singularity

A type of [spacetime singularity](#) (i.e. a boundary of [spacetime](#)) that is associated with infinitely high [energy density](#).

More information about the different types of singularities can be found in the spotlight text [Spacetime singularities](#).

ringdown

Just like a bell ‚rings‘ for some time after it’s been struck, the collision of two [black holes](#) resonates in the form of [gravitational waves](#). They are fainter than those released during the [merger](#) and occur due to a brief oscillation of the new black hole while it settles to a stationary configuration. This phase, during which gravitational waves are still being released, is called the ‚ringdown‘.

Schnelligkeit

In der Physik wird zwischen [Geschwindigkeit](#) und Schnelligkeit unterschieden. Die Geschwindigkeit ist eine gerichtete Größe (ein [Vektor](#)), sie enthält also neben einer Größeninformation auch eine Richtungsinformation.

Die Schnelligkeit ist der Betrag der Geschwindigkeit eines Objekts. Also zum Beispiel die Strecke, die das Objekt während einer bestimmten Zeitspanne zurücklegt, geteilt durch die Länge des Zeitintervalls. Wenn man das Zeitintervall unendlich klein macht, ist das Ergebnis die Schnelligkeit des Objekts zu einem bestimmten Zeitpunkt.

Schwarzschild black hole

A [solution](#) of [Einstein’s equations](#) found by Karl Schwarzschild in 1916, which corresponds to a model universe that contains a single, spherically symmetric [black hole](#).

More precisely, the Schwarzschild solution is a whole *family* of solutions: Schwarzschild’s formulae contain a free parameter m corresponding to the mass of the black hole. To each concrete value of m corresponds one specific solution to Einstein’s equations, a spacetime containing a spherically symmetric black hole of mass m .

The Schwarzschild solution is of practical importance as the outlying regions of the corresponding model universe describe the spacetime distortion around all kinds of objects that are spherically symmetric, or nearly so, such as the [sun](#) or the [earth](#) (cf. [Birkhoff’s theorem](#)).

Synonyms: Schwarzschild solution

Schwarzschild radius

A measure for the size of a spherically symmetric [black hole](#). It is defined using the area of the black hole's [horizon](#): In usual high school geometry (the geometry of [flat](#) space), radius and area of a spherical surface are related as

$$\text{area} = 4 \text{ times pi times radius}^2.$$

The Schwarzschild radius is defined indirectly by the formula

$$\text{Area of the black hole's horizon} = 4 \text{ times pi times (Schwarzschild radius)}^2.$$

It is directly proportional to the black hole's mass. The Schwarzschild radius for an object the mass of the [earth](#) is 9 millimetres, for an object with the [mass of the sun](#), 2.95 kilometres.

There is a quite general result that says: If a sphere of matter is compressed further and further, a black hole forms as soon as the sphere's radius gets smaller than the Schwarzschild radius corresponding to the matter's mass.

Schwarzschild solution

See [Schwarzschild black hole](#)

second

In the [International System of units](#): the basic unit of time. Defined as a certain multiple of the oscillation period of [electromagnetic radiation](#) set free in a certain transition within the [electron](#) shell of [atoms](#) of the type Cesium-133.

second of arc

See [arcminute](#), [arcsecond](#).

self-force, gravitational

Gravitational self-force is introduced in [black hole perturbation theory](#) in order to describe the effect of a particle's own [gravitational field](#) on its motion.

Shapiro delay

Also: gravitational time delay. In [general relativity](#), not only are light rays [deflected](#), in addition gravity can lead to light taking more time in its travels through space than in [classical](#) physics. This is called Shapiro effect or Shapiro delay. It has been measured numerous times for light signals in the solar system, for instance for radar waves sent from Earth to Venus and reflected back. These radar signals took measurably longer when their path led them closely by the massive sun.

Measuring this time delay is sometimes referred to as the “fourth test of general relativity”, in addition to the three [classical tests](#) of that theory.

Synonyms: Shapiro effect

shear

Generically, an extended body in [free fall](#) will experience deformation due to [tidal effects](#). For instance, in a body falling towards Earth, those parts that are slightly closer to the Earth will experience a slightly stronger gravitational pull than parts which are further away. Some of the deformation will change the body's volume. The shear is that part of the deformation which does *not* change the volume, only the body's shape.

Some examples for shear can be found in the spotlight text [Of singularities and breadmaking](#).

SI

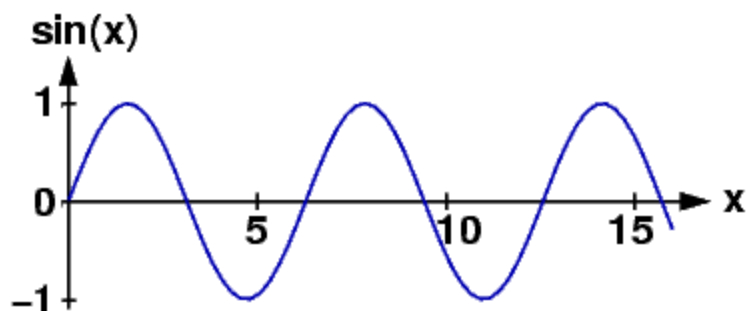
The international system of physical units, introduced in 1960. It is based on seven fundamental units; in the context of Einstein Online, the interesting ones are the [metre](#) as a measure of length and distance, the [second](#) as the unit of time, the [kilogram](#) as the unit of [mass](#) and the [Kelvin](#) as the unit of [temperature](#).

By multiplication and division, the seven fundamental units can be used to construct derived units for all other physical quantities. For instance, the unit of [speed](#) is the distance unit divided by the unit of time, metre per second.

Synonyms: *Système International d'Unités* International System of Units

sine

The sine, written $\sin(x)$, is a mathematical function that is perfectly regular and repetitive, with maximal and minimal values following each other in endless procession. The function is plotted here:



Sine waves are the simplest waves imaginable, with crests and valleys following each other in exactly the way described by a sine function.

singularity

Irregular boundary of [spacetime](#) in [general relativity](#) – region where spacetime simply comes to an end. Often, such boundaries are associated with spacetime [curvature](#) growing beyond all bounds and becoming infinitely large – so-called [curvature singularities](#) (notably [Ricci singularities](#) or [Weyl singularities](#) – but there are exceptions (for instance a [conic singularities](#)).

According to general relativity, a singularity exists inside every [black hole](#), and the starting point of any universe described by a [big bang model](#) is a singularity as well. The occurrence of singularities is a failure of general relativity – and a strong

indication that the theory is incomplete. Instead, one could describe the earliest universe and the interior of black holes using a theory of [quantum gravity](#).

More information about singularities can be found in the spotlight texts [Spacetime singularities](#) and [Of singularities and breadmaking](#).

Synonyms: spacetime singularity

singularity theorems

Theorems, proven by Roger Penrose and Stephen Hawking, that state under which circumstances [singularities](#) are inevitable in [general relativity](#). As the theorems assume the laws of general relativity and certain general properties of matter, but nothing else, they are valid quite generally. In particular, these theorems prove that, in the frame-work of general relativity, every [black hole](#) must contain a singularity, and every expanding universe like ours must have begun in a [big bang](#) singularity.

solar mass

The sun has a mass of $1.989 \cdot 10^{30}$ kilograms [see [exponential notation](#)] what is equal to 332 946 earth masses.

In astronomy, the solar mass is frequently used as a unit of mass (“Neutron stars typically have a mass of 1.4 solar masses”), sometimes written as $M_{\odot}$.

solar system

Our cosmic neighbourhood, consisting of a [star](#) – the [sun](#) -, eight [planets](#) orbiting the sun, numerous smaller bodies, dust and gas.

In the context of relativity, the solar system is interesting as a natural laboratory in which the prediction of [general relativity](#) can be tested – in particular those that differ from the predictions of [classical](#), [Newtonian gravity](#). Examples are the [relativistic perihelion shift](#) of planetary orbits, the [deflection of light](#) close to the sun and the [Shapiro effect](#).

solid state

[State of matter](#) in which the [atoms](#) or [molecules](#) are bound so tightly to each other so that they form a solid, stable lump. In contrast with [fluids](#), whose form adapts to whatever container they are placed in, solid bodies keep their form.

Compare the other [states of matter](#): [liquid](#), [gas](#), [plasma](#).

Synonyms: solid body

solution

In the context of [general relativity](#): a solution or, more precisely, a solution of the [Einstein equations](#) is a model universe that follows the law of gravity as prescribed by general relativity. See also [exact solution](#).

space

In a strict sense: Space as we know it from everyday life: the totality of all locations in which objects can sit, with three [dimensions](#).

In a more general sense used by mathematicians, all kinds of sets of points are spaces – a line for instance, which has but a single dimension, or a two-dimensional surface, but also higher-dimensional spaces. Also, in such more general spaces, geometry can be different from the standard [Euclidean geometry](#) taught in high schools – such spaces can be [curved](#).

Space Telescope Science Institute

The institute operating the [Hubble space telescope](#); located in Baltimore, USA.

[Space Telescope Science Institute website](#)

spacetime

Already in [special relativity](#), observers in motion relative to each other will not, in general, agree as to whether two events happen simultaneously, or as to how great is the distance between two objects. They do, however, agree as to what events there are, although not to when and where they happen. This observer-independent totality of all events is called spacetime. How spacetime is split into space and time can differ from observer to observer.

Every-day space has three [dimensions](#). Adding time adds another dimension – spacetime has four dimensions, all in all.

We are used to the idea of a point in space – an object that has only one location and is completely defined once its space coordinates are given. In spacetime, a spacetime point is an object defined completely once its space coordinates and its time coordinate are given – which makes a spacetime point nothing but an elementary event.

The idea of spacetime is, in addition to its role in special relativity, a building block of [general relativity](#). Analogous to how a [plane](#) is [flat](#), but the surface of a sphere is curved, in general relativity, curved or distorted versions of the simple, flat spacetime of special relativity play a role. Spacetime [curvature](#), in general relativity, is intimately connected with [gravity](#).

For an introduction to the basics of both theories of relativity, check out the chapters [Special relativity](#) and [General relativity](#) in [Elementary Einstein](#). Sometimes, it can be helpful to view spacetime in analogy to ordinary space – such analogies are explored in the spotlight topics [Time dilation on the road](#) (for [time dilation](#)) and [Twins on the road](#) (for the [twin effect](#)).

spacetime singularity

See entry [singularity](#) above.

special relativity

Albert Einstein's theory of the fundamentals of space, time, and movement (but not [gravity](#)). For a brief introduction, check out the chapter [Special relativity](#) of [Elementary Einstein](#).

Selected aspects of special relativity are described in the category [Special relativity](#) of our [Spotlights on relativity](#).

Synonyms: special theory of relativity

spectrum

The [electromagnetic radiation](#) reaching us from an astronomical object or other source is a mix of electromagnetic waves with a great variety of [frequencies](#). The spectrum lists the composition of this mix: For every frequency, it states the amount of radiation energy contributed by waves of that particular frequency.

Synonyms: spectra

speed

An object's average speed is the distance it moves during a given period of time, divided by the length of the time interval. If you make the time interval infinitely small, the result is the object's speed at one particular moment in time. The notion of speed can be applied to [waves](#) in different ways; for instance, for a simple wave, the *phase speed* is the speed at which any given wave crest or wave propagates through space.

See also the more general entry [velocity](#).

speed of light

The speed at which [light](#) or, more generally, [electromagnetic radiation](#) propagates through space (especially: through empty space). Central quantity in [special relativity](#): There, the constancy of the speed of light is a basic postulate: every observer (more precisely: every [inertial observer](#)) that measures the speed of light in vacuum obtains the same constant value, $c=299,792,458$ metres per second.

Another important relativistic aspect of the speed of light is that it defines an absolute upper speed limit: In [special relativity](#), nothing can move faster than light, and information or influence at most be transmitted at light-speed. In [general relativity](#), the same law is in force locally: No object, no matter, no information can directly overtake or catch up with light (cf. [causality](#)).

Basic information about the role of light speed in special relativity can be found in the chapter [Special relativity](#) of [Elementary Einstein](#).

sphere

A spherical surface is a simple example for a [curved surface](#). It is easily pictured as a surface embedded in three-dimensional space: a spherical surface is the set of all points at a certain fixed distance from a given point (the point being the centre of the sphere).

Mathematically, a spherical surface can be described without recourse to three-dimensional space – when mathematicians talk of the geometry of such a surface, they (almost) always mean the “inner geometry”: Those properties of the surface noticeable to two-dimensional beings, living and working in that surface, capable of measuring distances and angles in it.

Sphere is regularly used as a synonym for spherical surface (instead of describing a solid, three-dimensional ball). And not only for the two-dimensional spherical surface described above, but also for its analogues in lower and higher [dimensions](#). A one-sphere, for instance, is the same as a circle, a two-sphere is the spherical surface defined above, a three-sphere its three-dimensional analogue.

spin

Fundamental quantum property of [elementary](#) as well as of compound particles. For elementary particles, the spin determines whether the particle is a [matter particle](#) (half-integer spin such as $1/2$, $3/2$, $5/2$ etc.) or a [force particle](#) (integer spin such as 0, 1, 2 etc.).

spin network

In [loop quantum gravity](#) (one of the candidates for a theory of [quantum gravity](#)), the underlying microscopic structure of [space](#) is a spin network – a graph consisting of lines and nodes where each line is assigned a label consisting of a half-integer number. Mathematically, that number is closely connected with the [spin](#) of [elementary particles](#).

More information about spin networks can be found in the spotlight topic [The fabric of space: spin networks](#).

standard acceleration

The [acceleration](#) imparted by the earth's [gravitation](#) to a body located on the earth's surface: If you raise such a body up a bit and let it fall, it will accelerate 9.81 metres per second squared, in other words: in every second, its speed will increase by 9.81 metres per second.

Standard acceleration, abbreviated as g, is often used as a measure for accelerations. For instance, an acceleration of 2 g corresponds to $2 \cdot 9.81 = 19.62 \text{ m/s}^2$.

standard model of cosmology

Another name for the [big bang models](#).

standard model of elementary particle physics

The current state of the art for describing the basic properties of matter and forces. The standard model theories are based on [special relativity](#) and [quantum theory](#) and they describe the behaviour of elementary matter particles such as [electrons](#), [neutrinos](#) and [quarks](#) as well as their [anti-particles](#). It also describes three quantum forces acting between them: [electromagnetism](#), the [weak nuclear force](#) and the [strong nuclear force](#). These forces act by the exchange of [force particles](#). There is one elementary force for which no such quantum description exists, and which is not part of the standard model: [gravity](#).

Synonyms: standard model of particle physics

star

A cosmic gas ball that is massive enough for pressure and temperature in its core to reach values where self-sustained nuclear [fusion](#) reactions set in. The energy set

free in these reactions makes stars into very bright sources of light and other forms of [electromagnetic radiation](#).

Once the nuclear fuel is exhausted, the star becomes a [white dwarf](#), a [neutron star](#) or a [black hole](#).

Synonyms: stars

states (of matter)

Depending on parameters like temperature or pressure, matter can exist in different states with greatly varying physical properties. The most important states of matter are the solid, liquid, gaseous and plasma states (the latter three forms of matter are also called fluid). At very low temperatures, atoms bind together to form a [solid body](#) with a definite shape. As temperature increases, many solid bodies melt, turning into a [liquid](#) or, at even higher temperatures, a [gas](#), an ensemble of [atoms](#) and/or [molecules](#) wildly scurrying back and forth. A further increase in temperature transforms matter into a [plasma](#). In this state atoms disintegrate into [atomic nuclei](#) and [electrons](#).

stationary

Roughly speaking, in relativity, a situation or a [spacetime](#) is stationary if there is no change over time.

To be more precise, one has to take into account that, in [general relativity](#), [time](#) can be defined in many different ways, all of them equally valid for formulating the laws of physics. This leads to a modified definition: A situation or a spacetime is stationary if it is possible to define time in a way so that there is no change of its properties over time – if you follow the properties of a given region of space over time, they will not change.

Stefan-Boltzmann law

One of the laws governing the properties of the simplest form of [thermal radiation](#) – that emitted by a [blackbody](#): The total [energy](#) emitted by such a body is proportional to the fourth power of its temperature (measured in [Kelvin](#)).

stellar black holes

Stellar black holes are [black holes](#) with between a few and a dozen [solar masses](#) that are formed when the core of a massive [star](#) collapses.

Basic information about black holes can be found in the chapter [Black holes & Co.](#) of [Elementary Einstein](#).

straight line

In the [plane](#), in three-dimensional everyday [space](#) or in more general [flat spaces](#): Any line that forms the shortest connection between two given points.

In the spacetime of [special relativity](#): The [world-line](#) of an object moving with constant speed on a path that is straight in space.

Synonyms: straight line in space straight line in spacetime

string theory

Candidate for a theory of [quantum gravity](#); a [quantum theory](#), where the fundamental building blocks are tiny, one-dimensional, oscillating entities, called strings.

A brief description can be found on the page [String theory](#) in the chapter [Relativity and the quantum](#) of [Elementary Einstein](#).

Synonyms: string, strings

strong force

One of the four fundamental forces in our universe (the others are [electromagnetism](#), [weak nuclear force](#) and [gravity](#)). The strong force binds the [quarks](#) to form compound particles such as [protons](#) and [neutrons](#). Indirectly, it is also responsible for holding together protons and neutrons in atomic [nuclei](#).

Synonyms: strong nuclear force

sun

The central (and most massive) body of our [solar system](#); the [star](#) closest to us; a ball of gas with a radius of ca. 700000 km (for comparison the radius of the earth: 12756 km) and a mass of $1.989 \cdot 10^{30}$ kilograms [see [exponential notation](#)].

In the interior of the sun, nuclear [fusion](#) processes run their course; they are responsible for the sun's impressive brightness.

supergravity

Class of models that generalize [Einstein's general theory of relativity](#) in a way that it satisfies the requirements of [supersymmetry](#).

Today, supergravity is of interest in the context of [string theory](#): In the limiting case of low energies (where low energies includes everything accessible with modern [particle accelerators](#)), the physics of string theories can be described with the help of certain supergravity models.

supermassive black holes

are [black holes](#) with masses of more than a million [solar masses](#). As far as we know, such holes can be found in the central regions of almost all [galaxies](#). Central black holes are the energy source for [radio galaxies](#) and other [active galactic nuclei](#).

Basic informations about black holes in general can be found in the chapter [Black holes & Co.](#) of [Elementary Einstein](#).

supernova

Highly energetic explosion that ends the life of stars with more than about ten [solar masses](#). In this explosion, the outer layers of the stars are ejected into space, while the core regions collapse to form a [neutron star](#) or even a [black hole](#).

superstring theory

Synonym: supersymmetric [string theory](#). String theory that satisfies the requirements of an abstract symmetry called [superymmetry](#). All models of string theory that are realistic candidates for a theory of [quantum gravity](#), are superstring theories.

supersymmetry

Abstract [symmetry](#) that some of the models of [particle physics](#) satisfy: in such models, for every species of particles, there is a partner species with the same mass. If the particles are matter particles ([fermions](#)), then the partner particles are [force particles](#) ([bosons](#)), and vice versa.

surface

Geometric [space](#) with two [dimensions](#). Examples include the [plane](#) or the surface of a [sphere](#).

surface gravity

The [acceleration](#) due to [gravity](#) which is experienced by an object resting on the surface of some solid body is called the body's surface gravity (as most of the solid bodies in question are shaped by gravity, the value for the surface gravity tends to be the same everywhere on the body's surface). For the [Earth](#), the surface gravity is 9.81 metres per square second, the so-called [standard acceleration](#).

symmetry

A situation has a symmetry if certain changes make no difference. For instance: a mirror-symmetric image that you view in a mirror looks the same. A perfect sphere looks the same, even if it is rotated around an arbitrary axis through its centre point ("spherical symmetry").

In [particle physics](#), there are more abstract symmetries, less readily pictured, such as [supersymmetry](#).

synchrotron

A [particle accelerator](#), in which particles are accelerated with the help of electric fields, while strong magnetic fields keep them on track (the fact that ever-stronger magnetic fields are needed as acceleration proceeds is a consequence of the fact that [relativistic mass](#) increases with speed.)

synchrotron radiation

[Electromagnetic radiation](#) produced when electrically [charged](#) particles (for instance, [electrons](#)) are made to follow a curved trajectory in a [particle accelerator](#), or when these particles undergo comparable accelerations in nature.

Due to effects that can be derived from [special relativity](#), synchrotron radiation is densely concentrated and very intense. These properties, together with the fact that it is very easy to produce synchrotron radiation with a clearly defined [frequency](#), make this type of radiation a valuable tool for research not only in basic physics, but also in biology and medicine.

When it was first discovered, synchrotron radiation was an (annoying!) side effect, observable at particle accelerators which were used for research into the basic properties of [elementary particles](#). Nowadays, there are many accelerators whose main purposes is the production of this radiation!

For a list of important European Synchrotrons, have a look at > [FIS-Landschaft](#), with BESSY II, PETRA III or the ESRF.

In the US, you can find the > [National Synchrotron Light Source II at the Brookhaven National Laboratory](#).

Syracuse University

Research university (enrolment ca. 20,000) in New York State. Research topics of the physics department include classical and [quantum gravity](#), [cosmology](#) and [gravitational waves](#).

[Theoretical Cosmology and Elementary Particle Physics at Syracuse University](#)
[Gravitational Wave Group at Syracuse University](#)

Système International d'Unités

See [SI, International System of Units](#)

Synonyms: International System of Units

temperature

In systems consisting of many particles, be they [solid bodies](#), [fluids](#) or [gases](#), the constituents are in constant, chaotic motion: the atoms in a solid crystal oscillate a bit, the molecules of a gas are in rapid, disordered motion, and so on. The average energy with which each constituent contributes to every part of the disorderly motion is the same, and it is called the temperature of the system. High average energy corresponds to high temperature – atoms vibrating wildly, gas molecules zipping around very fast -, low average energy to low temperature.

In a slightly different context, certain mixtures of [electromagnetic radiation](#) can be assigned a temperature (“radiation temperature”), a single parameter that completely defines the basic properties of the radiation (more precisely, its [spectrum](#)). It corresponds to the [thermal radiation](#) emitted by a hot body with precisely that temperature.

In physics, temperature is measured in [Kelvin](#), in everyday life, depending on the country, in [Fahrenheit](#) or [Celsius](#).

templates

When measuring [gravitational waves](#), researchers use so-called templates in order to separate real gravitational wave signals from background noise. For this purpose,

they calculate the waveform of gravitational waves of certain different origins, such as coalescing [black holes](#). Knowledge of the signals' shapes facilitates their identification in the detector data.

test particles

In the context of [gravity](#): body whose [mass](#) is so small, that it can be used to probe the gravitational influences of their bodies, as its own gravitational field is too small to affect or change the situation in any significant way.

Analogously, in [electromagnetism](#): small, charged body with so little [charge](#) that it can be used to explore the electromagnetic influence of other bodies without its presence affecting or changing the situation in any significant way.

tests of general relativity, classical

The first two tests of [general relativity](#) were the comparison between prediction and observation for the [perihelion advance](#) of the planet Mercury and for the [deflection of light](#) near the Sun. In 1959, measurements of the [gravitational redshift](#) provided an additional test. All three effects in questions were predicted by Einstein, and these and subsequent measurements are known as the classical tests of general relativity. Measurements of the [Shapiro time delay](#) are sometimes called the “fourth test of general relativity”.

More information about the deflection of light can be found in the spotlight text [The gravitational deflection of light](#), while the connection of this effect with one of the fundamental principles of general relativity is explored in [The equivalence principle and the deflection of light](#).

Synonyms: classical tests of general relativity

TeV

See [electron volt](#)

Synonyms: Tera-electronvolt

theory of relativity

See [relativity theory](#)

Synonyms: theories of relativity

thermal energy

The [energy](#) contained in the disordered motion of a body's constituents – for instance, the energy of the disorderly motion of the [atoms](#) or [molecules](#) of a [gas](#), or their oscillation in a [solid body](#). If one increases a body's thermal energy, one also raises its [temperature](#).

thermal motion

What we call heat in everyday life corresponds to disordered motion of the microscopically small constituents of matter (say, [atoms](#) or [molecules](#)) – one example being the chaotic dance of the molecules making up a [gas](#), another the oscillation of the molecules forming a [solid body](#). This disordered motion is called thermal motion, and its average [energy](#) defines a system's [temperature](#) (cf. the preceeding entry on [thermal energy](#)).

thermal radiation

In a narrow sense: synonym for [infrared radiation](#).

In a more general sense: The [electromagnetic radiation](#) emitted by every body with non-zero [temperature](#) due to the laws of thermodynamics. The properties of this radiation (in particular: its [spectrum](#)) depend on the body's [temperature](#) – in the simplest case, that of what is called a [blackbody](#), they even depend on nothing but the temperature and some universal constants.

For everyday temperatures, such as those of a hotplate, thermal radiation is emitted mainly in the form of [infrared radiation](#). At higher temperatures, significant amounts of visible light are emitted, as well: a hotplate that is very hot indeed looks dark red or even light red; molten metal looks yellow or even white. In more extreme situations, thermal radiation can have energies that are even higher – for instance, the gases in the [accretion discs](#) of [black holes](#) are so hot that they emit great amounts of thermal radiation in the [x-ray](#) region.

thermodynamic equilibrium

A physical system is in thermodynamic equilibrium if its [energy](#) is distributed evenly among all the different ways in which its components can move or vibrate – what physicists call the system's “degrees of freedom” – and there is no flow of energy within or at the boundaries of the system. The average energy per degree of freedom is a direct measure of the system's [temperature](#).

For example, for a [gas](#), the [kinetic energy](#) of all its many particles is, on average, the same.

The totality of all [electromagnetic fields](#) is a physical system as well. More information about how thermal equilibrium of a hot body and the electromagnetic fields leads to the emission of [thermal radiation](#) can be found in the spotlight text [Heat that meets the eye](#).

The situation is slightly more complicated in systems that allows transmutations – for instance a system consisting of particles of species A and particles of another species B, where A-particles can change into B-particles and the other way around, For such a system, equilibrium at a certain [temperature](#) implies definite values for the relative abundances of the different particle species – how many particles of species A there should be, on average, for each particle of another species B. Such equilibria are of great importance for the physics of the early universe, as described by the [big bang models](#).

third-generation gravitational-wave detector

Third-generation gravitational-wave detectors are supposed to measure [gravitational waves](#) with higher sensitivity than current detectors such as Advanced [LIGO](#) or Advanced [Virgo](#). Examples for such projects are the [Einstein telescope](#) and [Cosmic Explorer](#). More information is available in our related spotlight article: [Third-generation gravitational-wave detectors](#).

tidal effects

Idealized situations apart, the [gravitational](#) influences acting on an object depend on the object's position. Take two small objects in the neighbourhood of a massive body: If one of them is closer to the massive body, it will be subject to a stronger gravitational pull. All effects that can be traced back to this variation of gravitational influences from location to location are called tidal effects.

Whenever gravitation is regarded as a [force](#) (notably in [Newton's theory of gravity](#)), tidal effects are caused by minute force differences – differences in the strength and direction of the gravitational force at one point in space, as compared to a neighbouring point. These force differences, in turn, are called tidal forces.

The best-known example for tidal effects is the one responsible for their name: High tide and low tide at the sea-shore are caused by position-dependent variations of the gravitational force – very roughly speaking, the oceans on the side of the earth facing the moon are pulled towards that heavenly body more strongly than the solid globe of the earth, and that globe in turn feels a stronger pull than the oceans on the side facing away from the moon.

In the context of [general relativity](#), tidal forces are especially interesting where [singularities](#) are concerned – in fact, the theory predicts that regions near a singularity are dominated by very strong and rapidly changing tidal forces (for more information on this, see the spotlight text [Of singularities and breadmaking](#)).

Synonyms: tidal forces

time

It is a fact of life that not all events in our universe happen concurrently – instead, there is a certain order. Defining a time coordinate or defining time, the way physicists do it, is to define a prescription to associate with each event a number so as to reflect that order – if event B happens after event A, then the number associated with B should be larger than that associated with A. The first step of this definition is to construct a clock: Choose a simple process that repeats regularly. (What is “regular”? Luckily, in our universe, all elementary processes such as a swinging pendulum, the oscillations of atoms or of electronic circuits lead to the same concept of regularity.) As a second step, install a counter: A mechanism that, with every repetition of the chosen process, raises the count by one.

With this definition, one can at least assign a time (the numerical value of the counter) to events happening at location of the clock. For events at different locations, an additional definition is necessary: One needs to define simultaneity. After all, the statement that some far-away event A happens at 12 o'clock is the same as saying that event A and “our clock counter shows 12:00:00” are simultaneous. The how and why of defining simultaneity – a centre-piece of

Einstein's [special theory of relativity](#) – are described in the spotlight topic [Defining “now”](#).

With all these preparations, physicists can, in principle, assign a time coordinate value (“a time”) to any possible events, and describes how fast or how slow processes happen, compared to that time coordinate.

time dilation

In [special relativity](#): From the point of view of an observer (more precisely: an [inertial observer](#)), a moving clock goes slower than an identically built clock at rest. All other processes moving alongside the clock (for instance: everything happening aboard a rocket speeding by) are slowed down in an identical fashion.

Time dilation can be mutual: When two inertial observers speed past each other, each will find that the other's clocks go slower. Some aspects of this unfamiliar mutuality are explored in the spotlight topic [The dialectic of relativity](#); a geometric analogy is presented in [Time dilation on the road](#).

In [general relativity](#), there is the phenomenon of *gravitational time dilation*: Roughly speaking, clocks in the vicinity of a mass or other source of gravity run more slowly than clocks which are farther away. This phenomenon is closely related to the [gravitational redshift](#).

torus

A torus (pl. tori) is a [surface](#) shaped like that of a donut or bagel.

It is possible to define analogous geometric objects, all of them finite in extent and closed in upon themselves, with more [dimensions](#) than two. These are also known as tori; whenever it is necessary to indicate such an object's dimensionality, one can simply add a qualifier to the name: The donut surface is a two-torus (two-dimensional), its three-dimensional analogue is a three-torus, and so on.

transversal

A [wave](#) is called transversal if the effects associated with it (the [electric forces](#) associated with an [electromagnetic wave](#), or the space distortions caused by a [gravitational wave](#)) act only in directions perpendicular to the wave's direction of propagation. For gravitational waves, some more information about this property can be found in the spotlight text [The wave nature of gravitational waves](#).

Synonyms: transversality

Transversalwelle

Bei einer Transversalwelle erfolgt die Schwingung senkrecht zur Ausbreitungsrichtung. Bei einer [elektromagnetischen](#) Welle sind dies elektromagnetische Kräfte, bei einer [Gravitationswelle](#) die Deformation der [Raumzeit](#). Mehr Details zu dieser Eigenschaft von Gravitationswellen finden sich im Vertiefungsthema [Die Wellennatur der Gravitationswellen](#).

triangle

In a [plane](#) and other [flat space](#): Geometrical object consisting of three points (“vertices”) with three connecting [straight lines](#).

The definition can be made more general, in a way that applies to [curved](#) spaces, as well: A geometric object consisting of three points (“vertices”) connected by three segments of [geodesics](#).

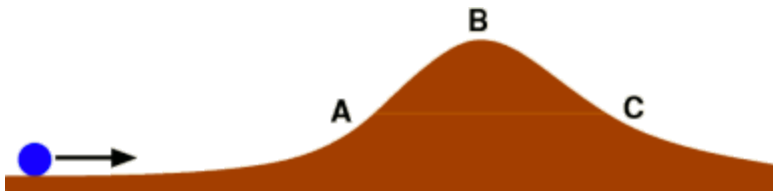
tritium

Variety of [hydrogen](#) in which the atomic [nucleus](#) contains two [neutrons](#) and a [proton](#). In ordinary hydrogen, the nucleus consists of a single proton; in heavy hydrogen ([deuterium](#)) there is one additional neutron.

In the context of [general relativity](#), tritium is of interest as one of the species of light atomic nuclei that formed in the early universe during [Big Bang Nucleosynthesis](#).

tunnel effect

A [quantum mechanical](#) phenomenon that can be pictured as follows. Imagine a ball rolling towards a hill:



Leaving quantum effects aside (in other words, in [classical](#) physics), we expect that what happens depends on the ball's [energy](#): If the ball moves fast enough (i.e. has sufficient energy), it will climb the hill, pass the peak at B and roll down on the other side. If the ball is too slow, it will reach some maximum height and then begin to roll back down without having passed B.

In the analogous situation for a [quantum particle](#), there is another possibility. Even an incoming particle with enough energy to climb to the height A, but not to pass the peak B, can appear on the right-hand side of the hill at point C and continue onwards. Such a transition is called tunneling – it is as if the particle had taken a secret tunnel from A to C to avoid the forbidden peak around B and arrive directly at C.

More generally, tunneling describes any transition from a state A to a state C that a quantum system can make, but that is forbidden to analogous systems in [classical](#) physics, since there, getting from A to C would only be possible by passing through a forbidden state B.

Synonyms: tunnelling, tunneling

twin effect

Effect of [special relativity](#), variant of the [time dilation effect](#): A twin that uses a high-powered rocket to travel in space with a speed near that of light before returning ages less than his twin sibling that has remained on Earth.

The question why this is sometimes thought to be a paradox, while it really isn't, is explored in the spotlight topic [The case of the travelling twins](#); a geometric analogy that is not about time, but about distance, is developed in [Twins on the road](#).

Synonyms: twin paradox

ultraviolet

Variety of [electromagnetic radiation](#) with [frequencies](#) between a few and a few hundred quadrillion oscillations per second, corresponding to [wave-lengths](#) between a few hundred and a few billionths of [metres](#). Known in everyday life as that part of the radiation we receive from the [sun](#) that causes our skin to tan

Synonyms: ultraviolet radiation

uncertainty principle

See [Heisenberg's uncertainty principle](#)

unified field theory

Collective designation for Einstein's unsuccessful attempts to formulate a theory in which [gravity](#) and other interactions, notably [electromagnetism](#), are described in a unified manner – a theory in which gravity and electromagnetism would be no more than different facets of one and the same underlying structure, in the same manner in which [magnetism](#) and the [electrostatic force](#) are facets of a more general description of electromagnetism.

After Einstein, quite a number of scientists have searched for a unified description of all interactions; the best-known modern incarnation of the idea of unification is [string theory](#).

uniqueness

Given a set of physical laws, one interesting class of question is aimed at finding out the variety of situations those laws allow. For example, is there only a single kind of rotating [black hole](#), or do the laws of [general relativity](#) admit an infinite variety of such objects? Theorems addressing this kind of question are generally known as uniqueness theorems – in their purest form, they state that, given a certain set of physical laws and a certain set of additional conditions, there is no more than one configuration of spacetime and matter that fits the bill.

In [general relativity](#), the most famous such theorems are the [black hole uniqueness theorems](#). They are explored in the spotlight text [How many different kinds of black hole are there?](#) A different aspect of the question of uniqueness is addressed in the spotlight text [The many ways of building an empty, unchanging universe](#).

Synonyms: uniqueness theorems

UV radiation

See [ultraviolet radiation](#).

velocity

In physics, velocity is a combination of two aspects: First of all, how fast an object is, in other words: how long a distance it moves in a given time (its “speed”). Secondly, the direction in which an object moves. Physicists combine these two informations into a single mathematical object, called a “vector”, and this is what is called the velocity. For instance, when a car goes around a curve with 100 miles per hour, its speed is constant, but as it changes its direction of movement, its velocity changes correspondingly.

Virgo

Virgo is a [gravitational wave detector](#) that was collaboratively built by French and Italian labs and was in operation until 2011. Then, the instrument was decommissioned and adapted to achieve 10 times higher sensitivity. Advanced Virgo started operation in 2017. Together with LIGO, GEO600 and KAGRA, it has since served the purpose of investigating space using gravitational wave signals.

Virgo Cluster

The [galaxy cluster](#) closest to our own, roughly 50 million [light years](#) away. It consists of about 2000 [galaxies](#). In the night sky, as viewed from Earth, it is located in the constellation virgo.

Synonyms: virgo galaxy cluster

visible light

In astronomy, the word [light](#) is often used to denote any kind of [electromagnetic radiation](#), from [infrared radiation](#) to [X-rays](#) and beyond. If, in contrast, an astronomer is talking about the ordinary light to which our eyes are susceptible – electromagnetic radiation with [wavelengths](#) between about 400 and 700 nanometers (400 to 700 billionths of a [metre](#)) – he or she will use the expression “visible light”.

wave

In a general sense: any travelling pattern, whether or not it involves matter being transported as well. Simple examples are water-waves – wave crests and troughs travelling over a water surface, and a Mexican wave in a football stadium, with fans alternately standing up and sitting down – the pattern moves through the stadium, not the fans themselves.

An especially simple form for a wave is a sinus wave, a regular pattern of wave crests and troughs.

Synonyms: waves

wavelength

For simple waves, where maxima and minima – wave crests and troughs – follow each other with perfect regularity, one can define a characteristic wavelength: the never-changing distance between two subsequent wave crests.

weak force

One of the basic interactions in the [standard model of particle physics](#). It is responsible for certain radioactive decays, such as the one where a [neutron](#) is transformed into a [proton](#), sending out an [electron](#) and a [neutrino](#).

Synonyms: weak nuclear force weak interaction

weightlessness

From everyday life, we're used to the Earth's gravity pulling every body down to the ground, and the strength of that force is called its weight. If no such force is present, then bodies placed at a certain location in space simply stay where they are, even without any support. Whenever that is the case, we are in a situation of weightlessness.

There are two types of situation in which weightlessness occurs. First of all, one could move far, far away into space, distancing oneself from all massive bodies so far that their gravitational influence becomes negligible. This would be a truly gravity-free situation. The second type of situation is more common. In free fall – be it an elevator plunging to earth, be it a space-station like the [ISS](#) in orbit around the Earth – bodies are weightless. The fact that, at least locally, there is no way to distinguish between these two types of weightlessness is embodied in the [equivalence principle](#), one of the fundamental building blocks of the [general theory of relativity](#).

Weyl singularity

A type of [spacetime singularity](#) (i.e. a boundary of [spacetime](#)) that is associated with infinitely strong [tidal forces](#).

More information about the different types of singularities can be found in the spotlight text [Spacetime singularities](#).

White dwarf

When [stars](#) with up to ten [solar masses](#) have exhausted the fuel of light atomic [nuclei](#) they need to sustain [nuclear fusion](#) reactions, they collapse to form a White Dwarf: a comparatively small ball of gas, prevented from further collapse by a quantum mechanical phenomenon, the so-called [degeneracy pressure](#) of its [electrons](#).

One way of determining the masses of White Dwarfs uses an effect of [general relativity](#), namely the [gravitational redshift](#) – more information about this can be found in the spotlight text [Gravitational redshift and White Dwarf stars](#).

Synonyms: White dwarf star

Why now? puzzle

From astronomical observations, it follows that the [density](#) associated with [dark energy](#) in our universe has the same order of magnitude as the density of the [matter](#) content of the universe. That is remarkable – in the past, the matter density will have been much larger than that of the dark energy, and in the far future, the roles

will be reversed. Is it coincidence that we make our observations precisely at the time when the two densities are of comparable size, or is there a physical explanation for it?

Synonyms: "Why now?" puzzle

Wien's law

One of the laws governing the properties of the simplest form of [thermal radiation](#) – that emitted by a [blackbody](#): the product of such a body's temperature (measured in [Kelvin](#)) and the [wavelength](#) at which it emits maximal amounts of energy is a universal constant.

Wilkinson Microwave Anisotropy Probe

[NASA](#) satellite telescope dedicated to observations of the [cosmic background radiation](#).

[WMAP website](#)

Synonyms: WMAP

world-line

The path of a pointlike object in four-dimensional [spacetime](#) is a line called the object's world-line. To every moment in time corresponds one point of that world-line giving the position of the object in space at that particular moment.

X-ray astronomy

Branch of physics devoted to the observation of [X-rays](#) reaching us from the depths of space. Such radiation is typically produced as the [thermal radiation](#) of matter at extremely high temperatures. An important example are the hot gases in the [accretion disc](#) of a [black hole](#).

X-rays

Highly energetic [electromagnetic waves](#) with [frequencies](#) between a few hundred Quadrillions and a few hundred Quintillions of oscillations per second, corresponding to [wavelengths](#) of a few billionths to a few trillionths of a metre. Most people know of these ray's medical applications – with their help, images of the interior of human bodies can be taken – however, they are also used in [\(X-ray\) astronomy](#).

XMM Newton

Satellite project of the European space agency [ESA](#). Launched in December 1999, XMM Newton is a space-based [X-ray](#) telescope; as such it is especially suited for research on the luminous phenomena associated with [black holes](#).

[XMM-Newton Pages at NASA](#)

[XMM-Newton Science Operations Centre Home Page \(ESA\)](#)

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